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RESEARCH ARTICLE

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On smooth rationally connected projective threefolds of Picard number two admitting int-amplified endomorphisms

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Dedicated to Professor Yongnam Lee on the occasion of his 60th birthday.

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Abstract

We prove that a smooth rationally connected projective threefold of Picard number two is toric if and only if it admits an int-amplified endomorphism, thereby answering Fakhruddin's question affirmatively in dimension three and Picard number two. As a corollary, we show that a blowup-equivariant smooth curve for a nonisomorphic surjective endomorphism of $\mathbb{P}^3$ must be a line.

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1 | INTRODUCTION

We work over an algebraically closed field $\mathbf{k}$ of characteristic zero.

As one of the most classical problems in algebraic geometry, the classification of smooth projective varieties X admitting nonisomorphic surjective endomorphisms f has seen a lot of progress in recent years. When $\dim X = 1$, the Hurwitz formula implies that X is either a rational or an elliptic curve. In the case $\dim X = 2$, the structure of such (possibly singular) varieties has been completely described by Nakayama (see [45]).

In higher dimensions, on the one hand, to exclude the nonessential case when an endomorphism is simply a product of some endomorphism with an automorphism, it is necessary to impose a constraint on f. The appropriate condition, introduced by the second author, turns out to be *int-amplified*. A surjective endomorphism $f : X \to X$ of a normal projective variety X is said to be int-amplified if one of the following equivalent conditions holds (see [33, Theorem 1.1] and [42, Proposition 3.7]).

- There exists an ample Cartier divisor L on X such that $f^*L - L$ is ample.
- Every eigenvalue of $f^*|_{\mathrm{NS}(X)}$ has modulus greater than one.
- The last dynamical degree of f is strictly greater than the other dynamical degrees of f; in this case, f is also said to be $(\dim X)$-*cohomologically hyperbolic*.

Int-amplified endomorphisms have emerged as powerful tools in the study of general endomorphisms, particularly in reducing a nonisomorphic endomorphism to a *polarized* one, that is, an endomorphism f satisfying $f^*H \sim qH$ for some ample divisor H and for some integer $q > 1$. For concrete applications in various contexts, we refer the reader to [26, 31, 40, 43, 52] and the references therein.

On the other hand, the three canonical fibrations, namely, the Kodaira fibration, the Albanese map (or more specifically, the Beauville–Bogomolov–Yau decomposition), and the maximal rationally connected fibration (or more specifically, the Cao–Höring decomposition), naturally facilitate the decomposition of endomorphisms via equivariant descent. This framework

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highlights rationally connected varieties as one of the central building blocks in the study of endomorphisms of higher dimensional varieties.

In light of the aspects introduced, it becomes a fundamental problem to characterize smooth rationally connected projective varieties admitting int-amplified endomorphisms. In this direction, generalizing the work of Fakhruddin [12, Question 4.4], the last two authors proposed [41, Question 1.2], stated as follows.

Question 1.1 (Fakhruddin). Let X be a smooth rationally connected projective variety admitting an int-amplified endomorphism f. Is X a toric variety?

Nakayama [44, Conjecture 2] confirmed Sato's conjecture that a smooth projective rational surface admitting a nonisomorphic surjective endomorphism is toric, in particular, Question 1.1 has a positive answer in dimension two. Question 1.1 also extends the following long-standing Conjecture 1.2 of the 1980s by noting that a smooth rationally connected variety of Picard number one is automatically Fano.

Conjecture 1.2. *Let X be a smooth Fano variety of Picard number one. Suppose that X admits a nonisomorphic surjective endomorphism f. Then X is a projective space.*

Conjecture 1.2 is known for threefolds [2, 20], for rational homogeneous spaces [47], and for hypersurfaces [5]. We refer the reader to [31, 50, 51] and the references therein for more information and recent progress on Conjecture 1.2.

Building on the works [2, 20], as well as the theory of the *equivariant minimal model program* (EMMP) for polarized and int-amplified endomorphisms, initiated by Zhang and the second author in [33, 36, 38], Zhang and the last two authors established an affirmative answer to Question 1.1 for Fano threefolds [43].

In this paper, we take a first step toward addressing Question 1.1 for arbitrary smooth rationally connected projective threefolds. Our main result is stated as follows.

Theorem 1.3. *A smooth rationally connected projective threefold X of Picard number 2 is toric if and only if it admits an int-amplified endomorphism.*

Proof. The only if part is clear by [44, Lemma 4] and [41, Proof of Theorem 1.4]. Now we deal with the if part. By Theorem 2.4, the threefold X admitting an int-amplified endomorphism is of Fano type. Say (X, Δ) is log Fano. Applying the cone Theorem [28, Theorem 3.7] to (X, Δ), we have two contractions $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}$ as in Notation 3.1. By Lemma 3.2, it suffices to treat the following situations.

- Case $\mathcal{S}_{\mathrm{Fano}}$. Given $\dim X = 3$ and $\rho(X) = 2$, an extremal Fano contraction of X has target either a surface or a curve. This yields two subcases:
  - Theorem 4.1: X admits a Fano contraction to a surface;
  - Theorem 6.1: X admits a Fano contraction to a curve;
- Case $\mathcal{S}_{\mathrm{Bir}}$. According to [35, Theorem 3.3], among the birational contractions of X, at least one is divisorial. This yields two subcases:
  - Theorem 7.1: X admits two divisorial contractions;
  - Theorem 8.1: X admits one divisorial contraction and one small contraction. □

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Remark 1.4. In fact, the last two cases in the proof of Theorem 1.3 do not occur and thus such a threefold has to be either a splitting $\mathbb{P}^1$-bundle over $\mathbb{P}^2$ or a splitting $\mathbb{P}^2$-bundle over $\mathbb{P}^1$. In general, any smooth projective toric variety of Picard number two is a splitting projective bundle over a projective space, see [3, Corollary 4.4].

Main difficulties. Previously in [43], the analysis relied heavily on the existence of two K_X-negative extremal contractions for Fano threefolds. Notably, the case of small contractions did not arise, thanks to Mori's seminal work [35, Theorem 3.3]. In contrast, in the present paper, only one K_X-negative extremal contraction is guaranteed, and the second extremal contraction may be small, in which case, the classification of Fano type threefolds as well as their contractions is not available. Even when the second contraction is log divisorial, little is known about its structure.

Furthermore, for Fano threefolds in [43], when Fano contractions occur, we only need to deal with $\mathbb{P}^1$-fibrations. In this paper, however, we must also handle more intricate cases, including quadric surface fibrations and sextic del Pezzo fibrations.

In what follows, we give an interesting application of Theorem 1.3 to the following significant and challenging Conjecture 1.5. The case of $\mathbb{P}^2$ was completely solved due to Gurjar [19] or Fornæss–Sibony [15, section 4]. In the case of $\mathbb{P}^3$, it is known that Y is linear if it is a prime divisor, thanks to the works of Nakayama–Zhang [46, Theorem 1.5] and Höring [21, Corollary 1.2]. For higher dimensional projective spaces $\mathbb{P}^n$, the Y is known to be linear if it is a smooth (or even with isolated singularities) hypersurface, by results of Beauville [5, Theorem], Cerveau–Lins Neto [8], Paranjape–Srinivas [47, Proposition 8], and Mabed [32, Corollary 1.4]. Note that all these known results address only the case where Y is a prime divisor, as the proofs primarily rely on properties of the sheaf of logarithmic differential forms. For higher codimensional subvarieties, this conjecture remains intractable at present.

Conjecture 1.5. *Let f be a surjective endomorphism of $X := \mathbb{P}^n$ with $\deg f \geqslant 2$. Then any f^{-1}-invariant (or f^{-1}-periodic) closed subvariety Y is a linear subspace of X.*

Let $Y \subsetneq X = \mathbb{P}^n$ be a smooth closed subvariety. We say Y is *blowup-equivariant* with respect to a nonisomorphic surjective endomorphism $f : X \to X$ if there exists a surjective endomorphism $\widehat{f} : \widehat{X} \to \widehat{X}$ such that $f \circ \pi = \pi \circ \widehat{f}$. As an application of Theorem 1.3, we show the linearity of a blowup-equivariant smooth curve in $\mathbb{P}^3$.

Corollary 1.6. *Let $f : X = \mathbb{P}^3 \to X$ be a surjective endomorphism with $\deg f \geqslant 2$. Let $Y \subsetneq \mathbb{P}^3$ be a smooth curve such that the blowup $\pi : \mathrm{Bl}_Y X \to X$ is f-equivariant. Then Y is a line.*

Proof. Note that $X' := \mathrm{Bl}_Y X$ has Picard number 2, f lifts to a surjective endomorphism $f' : X' \to X'$ by assumption, and f' is also polarized by [36, Corollary 3.12]. By Theorem 1.3, we see that X' is toric. Let T be the big torus acting on X'. Then π is also T-equivariant; see [7, Proposition 2.1]. Note that T is a maximal torus of the projective linear group $\mathrm{PGL}(4, \mathbf{k})$. So, after changing coordinates, we may assume that T is the diagonal subgroup of $\mathrm{PGL}(4, \mathbf{k})$ by [22, Corollary A, p. 135]. Clearly, T-invariant closed subvarieties of $\mathbb{P}^3$ must be linear. □

We conclude this section with the following example. Motivated by this example, we expect in general that an f^{-1}-invariant curve in $\mathbb{P}^n$ with $\deg f \geqslant 2$ is blowup-equivariant up to a deformation of f.

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Example 1.7. Consider the following family of endomorphisms parameterized by t.

$$f_t : \mathbb{P}^3 \longrightarrow \mathbb{P}^3$$
$$(x_0 : x_1 : x_2 : x_3) \longmapsto \left(x_0^2 + tx_1x_2 : x_1^2 : x_2^2 : x_3^2\right)$$

First, we assume that $t \neq 0$. Let L_{ij} be the line defined by $x_i = x_j = 0$. Then it is straightforward to see that L_{13} and L_{01} are f_t^{-1}-invariant. We claim that $\mathrm{Bl}_{L_{13}}\mathbb{P}^3 \to \mathbb{P}^3$ is f-equivariant but $\mathrm{Bl}_{L_{01}}\mathbb{P}^3 \to \mathbb{P}^3$ is not. Note that the blowup $\mathrm{Bl}_{L_{ij}}\mathbb{P}^3$ is the graph of the natural projection $\mathbb{P}^3 \dashrightarrow \mathbb{P}^1$ via $(x_0 : x_1 : x_2 : x_3) \mapsto (x_i : x_j)$. Hence, the blowup-equivariance is equivalent to the descending $f_t|_{\mathbb{P}^1}$ being well-defined. For the former case, it is clear that f_t induces a well-defined endomorphism $g_t : \mathbb{P}^1 \to \mathbb{P}^1$ with $(x_1 : x_3) \mapsto (x_1^2 : x_3^2)$. For the case L_{01}, the equivariance forces $g_t(x_0 : x_1) = (x_0^2 + tx_1x_2 : x_1^2)$, which is not well-defined when $t \neq 0$. Now, if we consider the endomorphism f_0, then both blowups are f_0-equivariant with a similar argument.

2 | PRELIMINARIES

2.1 | Varieties, divisors, and endomorphisms

Notation 2.1. Let X be a normal projective variety. We use the following notation throughout this paper unless otherwise stated.

$\mathrm{Pic}(X)$	the group of Cartier divisors of X modulo linear equivalence $\sim$
$\mathrm{Pic}^0(X)$	the neutral connected component of $\mathrm{Pic}(X)$
$\mathrm{NS}(X)$	$\mathrm{Pic}(X)/\mathrm{Pic}^0(X)$, the Néron–Severi group
$\mathrm{N}^1(X)$	$\mathrm{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$, the space of $\mathbb{R}$-Cartier divisors modulo numerical equivalence $\equiv$
$\mathrm{N}_{n-1}(X)$	the space of Weil $\mathbb{R}$-divisors modulo weak numerical equivalence $\equiv_w$
$\mathrm{Nef}(X)$	the cone of nef classes in $\mathrm{N}^1(X)$
$\mathrm{PE}(X)$	the cone of pseudo-effective classes in $\mathrm{N}^1(X)$
$\mathrm{PE}_{n-1}(X)$	the cone of pseudo-effective Weil $\mathbb{R}$-divisors in $\mathrm{N}_{n-1}(X)$
$\mathrm{Supp}\, D$	the support of effective divisor $D = \sum a_i D_i$ which is $\bigcup_{a_i>0} D_i$, where the D_i are prime divisors
$\rho(X)$	Picard number of X which is $\dim_{\mathbb{R}} \mathrm{N}^1(X)$

Here, two Weil $\mathbb{R}$-divisors P_1 and P_2 are said to be *weakly numerically equivalent* (denoted by $P_1 \equiv_w P_2$) if for any Cartier divisors $H_1, \ldots, H_{n-1}$, one has $(P_1 - P_2) \cdot H_1 \cdots H_{n-1} = 0$. Note that two $\mathbb{R}$-Cartier divisors $D_1 \equiv D_2$ if and only if $D_1 \equiv_w D_2$; see [36, Definition 2.2 and Lemma 2.3].

Definition 2.2. Throughout this article, the varieties below will be heavily involved.

(1) A normal projective variety X is of *Fano type*, if there is an effective Weil $\mathbb{Q}$-divisor Δ on X such that the pair (X, Δ) has at worst klt singularities and $-(K_X + \Delta)$ is ample and $\mathbb{Q}$-Cartier; the pair (X, Δ) is called *log Fano*. If $\Delta = 0$, then X is the usual klt Fano variety. We say that X is *weak Fano* if X is of Fano type and $-K_X$ is further nef (and hence big).
(2) A normal projective variety X is *of Calabi–Yau type* (resp., *of klt Calabi–Yau type*), if there is an effective Weil $\mathbb{Q}$-divisor Δ on X such that the pair (X, Δ) has at worst lc (resp., klt)

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singularities and $K_X + \Delta$ is $\mathbb{Q}$-Cartier and numerically trivial; the pair (X, Δ) is called *log Calabi–Yau* (resp., *klt log Calabi–Yau*).

(3) A normal variety X of dimension n is a *toric variety* if X contains a *big torus* $T = (\mathbf{k}^*)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on the whole variety; in this case, the complement $B := X \backslash T$ is a divisor and the pair (X, B) is said to be a *toric pair* with $K_X + B \sim 0$.

Definition 2.3. Let $f : X \to X$ be a surjective endomorphism of a projective variety X.

(1) We say that f is *q-polarized* if $f^*H \sim qH$ for some ample Cartier divisor H and integer $q > 1$, or equivalently, $f^*|_{\mathrm{N}^1(X)}$ is diagonalizable with all the eigenvalues being of modulus q [36, Proposition 2.9].

(2) We say that f is *int-amplified* if $f^*L - L$ is ample for some ample Cartier divisor L, or equivalently, every eigenvalue of $f^*|_{\mathrm{N}^1(X)}$ is of modulus greater than 1 [33, Theorem 1.1]. Clearly, every polarized endomorphism is int-amplified.

(3) A subset $S \subseteq X$ is called f^{-1}-*invariant* (resp., f^{-1}-*periodic*) if $f^{-1}(S) = S$ (resp., $f^{-n}(S) = S$ for some $n > 0$).

By implementing the EMMP, Yoshikawa has made significant advancements toward a solution to Question 1.1.

Theorem 2.4 [52, Corollary 1.4]. *A smooth rationally connected projective variety is of Fano type if it admits an int-amplified endomorphism.*

2.2 | Singularities and positivities

In this subsection, we study the intersection of boundaries for a given log canonical pair in the fiber spaces. Together with the canonical bundle formula and the dynamical assumption, we obtain several positivities of (relative) anti-log canonical divisors. We begin with the following definition.

Definition 2.5.

(1) Let $f : X \to Y$ be a finite surjective morphism between normal varieties. The *ramification divisor* R_f of f is defined by the formula $K_X = f^*K_Y + R_f$. We denote by $B_f = f(\mathrm{Supp}\, R_f)$ the (reduced) *branch divisor* of f.

(2) Let $\pi : P \to Y$ be a surjective projective morphism with Y being normal. Let $\nu : P^n \to P$ be the normalization, and $P^n \xrightarrow{\phi} \widetilde{Y} \xrightarrow{\sigma} Y$ the Stein factorization of $\pi \circ \nu$. Then we define the *general branch divisor of* π by $B_\pi := B_\sigma$.

When the base is a curve, the following lemma is an easy observation.

Lemma 2.6. *Let $\pi : P \to Y$ be a surjective projective morphism of varieties with Y being smooth and* $\dim Y = 1$. *Let $y \in B_\pi$. Suppose P is smooth at generic points of $\pi^{-1}(y)$. Then π^*y is not a reduced divisor.*

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Proof. Suppose the contrary that π^*y is a reduced divisor. Let $\nu : P^n \to P$ be the normalization. Let $P^n \xrightarrow{\phi} \widetilde{Y} \xrightarrow{\sigma} Y$ be the Stein factorization of $\pi\circ\nu$. By the assumption, ν is isomorphic at generic points of $\pi^{-1}(y)$. Then $\phi^*\sigma^*y = \nu^*\pi^*y$ is also a reduced divisor. Note that $\widetilde{Y}$ is a smooth curve. Then σ^*y is reduced, contradicting the choice of y. □

Corollary 2.7. *Let $\pi : X \to Y$ be a projective morphism between smooth varieties with* $\dim Y = 1$. *Let P be a prime divisor on X dominating Y and let $y \in B_{\pi|_P}$. Then $(X, P + \pi^*y)$ is not lc.*

Proof. Suppose the contrary that $(X, P + \pi^*y)$ is lc. Then $\pi^*y = \pi^{-1}(y)$ is reduced. Since X is smooth, $P \cap \pi^{-1}(y)$ is of pure codimension 2. Then $P + \pi^{-1}(y)$ is a simple normal crossing divisor near any generic point η of $P \cap \pi^{-1}(y)$. In particular, P is smooth at generic points of $(\pi|_P)^{-1}(y)$ and $(\pi|_P)^*y$ is a reduced divisor on P. However, this contradicts Lemma 2.6. □

In contrast to Corollary 2.7, we extend [33, Theorem 1.6] to pair version: Lemma 2.8. Lemmas 2.9 and 2.10 are by-products of Lemma 2.8.

Lemma 2.8. *Let $f : X \to X$ be an int-amplified endomorphism on a normal projective variety X. Let Δ be a reduced effective f^{-1}-periodic Weil divisor. Then $-(K_X + \Delta)$ is a pseudo-effective Weil divisor. Suppose further $K_X + \Delta$ is $\mathbb{Q}$-Cartier. Then (X, Δ) is lc and $-(K_X + \Delta)$ is $\mathbb{Q}$-effective.*

Proof. After iteration, we may assume $f^{-1}(\Delta) = \Delta$. By the ramification divisor formula, we have

$$f^*(-(K_X + \Delta)) - (-(K_X + \Delta)) = R_f - (f^*\Delta - \Delta)$$

where the right-hand side is effective. By [33, Theorem 3.3], the divisor $-(K_X + \Delta)$ is pseudo-effective. It is further $\mathbb{Q}$-effective when $K_X + \Delta$ is $\mathbb{Q}$-Cartier by the same proof (after replacing K_X by $K_X + \Delta$) of [39, Theorem 6.2].

Suppose the contrary that (X, Δ) is not lc. Let Z be any irreducible component of the non-lc locus $\mathrm{Nlc}(X, \Delta)$ which is of dimension $d < \dim X$. By [6, Theorem 1.4] we have $\deg f|_Z = \deg f$ and then $f_*Z = (\deg f)Z$. Let H be an ample divisor of X. Then $(f^i)^*H^d \cdot Z = (\deg f)^i H^d \cdot Z$, which contradicts [33, Lemma 3.8]. □

Lemma 2.9 (cf. Lemma 8.11). *Let $f : X \to X$ be an int-amplified endomorphism of a smooth projective variety X. Suppose A and B are two different f^{-1}-periodic prime divisors. Then $A + B$ is a simple normal crossing divisor around the generic points of the intersection $A \cap B$. In particular, $A|_B$ and $B|_A$ are reduced divisors.*

Proof. By Lemma 2.8, the pair $(X, A + B)$ is lc. By Bertini's theorem, we can take a sufficiently general smooth hyperplane section H such that $(X, A + B + H)$ is lc. By [28, Theorem 5.50], the pair $(H, (A + B)|_H)$ is lc. Repeating this process $\dim X - 2$ times, we may assume that X is a smooth projective surface. For each point $p \in A \cap B$, it follows from [28, Lemma 2.29] that both A and B are smooth along the point p and the blowup along p will separate A and B. So, we have the transversality of $A \cap B$. □

Lemma 2.10. *Let $f : X \to X$ be an int-amplified endomorphism of a smooth projective variety X. Let D be a reduced f^{-1}-periodic divisor. Let $S \subseteq \mathrm{Sing}(D)$ be an irreducible closed subset with* $\dim S = \dim X - 2$. *Then S is f^{-1}-periodic and D is a normal crossing divisor around the generic point of S.*

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Proof. By Lemma 2.8, the pair (X, D) is lc. Computing the discrepancy [28, Lemma 2.29], we see that the Cartier divisor D has multiplicity exactly 2 at the generic point of S and S is a log canonical center of (X, D). It follows from [6, Corollary 3.3] that S is f^{-1}-periodic. By a similar proof of Lemma 2.9 and [28, Theorem 4.15], we see that D has exactly two analytic branches crossing transversally at the generic point of S. □

In the last part of this subsection, we show the following lemma, which extends [33, Theorem 1.5] to the relative version, and give an interesting application Corollary 2.12 to the canonical bundle formula.

Lemma 2.11. *Let $\pi : X \to Y$ be an equi-dimensional fibration of normal projective varieties whose fibers are all reduced and irreducible. Assume that we have the equivariant dynamical system*

$$f \circlearrowright X \xrightarrow{\pi} Y \circlearrowleft g.$$

where f, g are int-amplified endomorphisms. Let E be a reduced and horizontal divisor such that $f^{-1}(E) = E$. Then $-K_{X/Y} - E$ is a pseudo-effective Weil divisor.

Proof. Since π is equi-dimensional and Y is normal, it follows that the inverse image $\pi^{-1}(\operatorname{Sing} Y)$ of the singular locus of Y is of codimension $\geqslant 2$ in X. Then for any prime divisor Q on Y, we can define the pullback π^*Q first outside the singular locus (as the usual pullback of Cartier divisors) and then take the closure by the normality of X. Besides, for any prime divisor P on X, either P dominates Y, or $\pi(P)$ is a prime divisor on Y. So, for any divisor P on X, we can write $P = P^h + P^v$ where P^h is the summand, the support of whose irreducible component dominates Y and $P^v = P - P^h$ is the vertical part whose π-images are divisors on Y.

Let D be the Weil divisor $-(K_{X/Y} + E)$. By the ramification divisor formula, we have

$$K_X = f^*K_X + R_f,\ K_Y = g^*K_Y + R_g,\ R_f = R_f^h + R_f^v.$$

Then we obtain that

$$\begin{aligned} f^*D - D &= f^*(-(K_{X/Y} + E)) + (K_{X/Y} + E) \\ &= ((K_X + E) - f^*(K_X + E)) + (f^*\pi^*K_Y - \pi^*K_Y) \\ &= \left(R_f^h - f^*E + E\right) + \left(R_f^v - \pi^*R_g\right). \end{aligned}$$

For any prime divisor Q on Y, its pullback π^*Q is also a prime divisor because the fibers are all irreducible and reduced. This implies $\pi^*R_g \leqslant R_f^v$. In addition, by the effectivity of the logarithmic ramification divisor (see [23, Theorem 11.5]), we also have $R_f^h - f^*E + E \geqslant 0$. Consequently, $f^*D - D$ is effective. By [33, Theorem 3.3], we see that $-K_{X/Y} - E$ is a pseudo-effective Weil divisor. □

Lemma 2.11 fails if we drop the equi-dimensional assumption. For example, if we take a one-point blowup $X \to Y \cong \mathbb{P}^2$, then we can construct a polarized endomorphism f on X as in [44, Lemma 4]. In this case, $K_{X/Y}$ is a nonzero effective divisor.

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Corollary 2.12. *With the same notation as in Lemma 2.11, assume further that $K_X + E$ is π-trivial. Then $(X, E + \pi^*P)$ is lc for any prime divisor P on Y.*

Proof. Since (X, E) is lc and π-trivial, by the canonical bundle formula (see, e.g., [14, section 3]), we have

$$K_X + E \sim_{\mathbb{Q}} \pi^*(K_Y + M_Y + B)$$

where B is the (effective) discriminant divisor of (π, E) defined by $B = \sum_P (1 - b_P)P$ and M_Y is the b-nef (and hence pseudo-effective) $\mathbb{Q}$-divisor. Here, P is any prime divisor and b_P is given by

$$b_P = \max\{t \in \mathbb{Q} \mid (X, E + t\pi^*P) \text{ is sub lc over the generic point of } P\}.$$

By Lemma 2.11, $-K_{X/Y} - E$ is pseudo-effective and hence so is $-\pi^*(M_Y + B)$. Therefore, $B = 0$ and $M_Y \equiv 0$ which concludes our corollary. □

2.3 | Endomorphisms on some del Pezzo surfaces

We prepare some simple facts on the smooth quadric surface (which is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$) and the sextic del Pezzo surface (which is the blowup of $\mathbb{P}^2$ at three noncollinear points). The main task is to introduce the cross $+$ and hexagon $\hexagon$ in Notations 2.14 and 2.18 for the purpose of better demonstrating the proofs of Theorems 6.3 and 6.6.

Lemma 2.13. *Let $f : X = \mathbb{P}^1 \times \mathbb{P}^1 \to X$ be a surjective endomorphism. Then either $f = f_1 \times f_2$ or $f \circ \sigma = f_1 \times f_2$ where $\sigma(x_1, x_2) = (x_2, x_1)$.*

Proof. Let $(a, b) \in X$. Denote by $H_1 = \{a\} \times \mathbb{P}^1$ and $H_2 = \mathbb{P}^1 \times \{b\}$. Note that $\mathrm{Nef}(X) = \mathbb{R}_{\geqslant 0}H_1 + \mathbb{R}_{\geqslant 0}H_2$. Then either $f^*H_1 \equiv qH_1$ or $\sigma^*f^*H_1 \equiv qH_1$ for some $q > 0$. For the first case, $f^*H_2 \equiv q'H_2$, and the two projections of X are both f-equivariant. So, f splits. For the second case, the same argument holds for $f \circ \sigma$. □

Notation 2.14. For a point $(a, b) \in \mathbb{P}^1 \times \mathbb{P}^1$, denote by

$$+_{(a,b)} := \{a\} \times \mathbb{P}^1 \cup \mathbb{P}^1 \times \{b\}$$

the *cross* $+ \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ with (a, b) being the anchor point of the crossing.

For a subset $S \subseteq \mathbb{P}^1 \times \mathbb{P}^1$, denote by

$$+_S := \bigcup_{(a,b)\in S} +_{(a,b)}$$

the union of *crosses* anchored in S.

Lemma 2.15. *Let A be an ample Weil divisor on $\mathbb{P}^1 \times \mathbb{P}^1$. Suppose* $\mathrm{Supp}\, A \subseteq +_S$ *for some finite set S. Then* $\mathrm{Supp}\, A = +_{\mathrm{Sing}(\mathrm{Supp}\, A)}$.

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Proof. By the assumption, we have

$$\operatorname{Supp} A = \left(\bigcup_{i=1}^{m}\{a_i\}\times\mathbb{P}^1\right)\cup\left(\bigcup_{j=1}^{n}\mathbb{P}^1\times\{b_j\}\right)$$

where $m, n \geqslant 1$ by the ampleness of A. Then

$$\operatorname{Supp} A = +_T$$

where $T = \bigcup_{1\leqslant i\leqslant m, 1\leqslant j\leqslant n}\{(a_i, b_j)\}$. Finally, note that $\operatorname{Sing}(\operatorname{Supp} A) = T$. □

By Lemma 2.13, the following lemma is clear.

Lemma 2.16. *Let $f : X = \mathbb{P}^1\times\mathbb{P}^1 \to X$ be a surjective endomorphism. Let $S \subseteq X$. Then $f^{\pm}(+_S) = +_{f^{\pm}(S)}$.*

Lemma 2.17. *Let $f : X = \mathbb{P}^1\times\mathbb{P}^1 \to X$ be an int-amplified endomorphism. Let $S \subseteq X$ be a finite subset such that $\sharp S = \sharp f^{-1}(S)$. Then $+_S \subseteq B_f = +_{\operatorname{Sing}(B_f)}$.*

Proof. Let $\sigma(x_1, x_2) = (x_2, x_1)$. Then $\sharp(f\circ\sigma)^{-1}(S) = \sharp S$ and $B_{f\circ\sigma} = B_f$. So, we may assume $f = f_1\times f_2$ by Lemma 2.13. Note that $\deg f_1$ and $\deg f_2$ are both greater than 1 because f is int-amplified. Then $B_f = (B_{f_1}\times\mathbb{P}^1)\cup(\mathbb{P}^1\times B_{f_2}) = +_{\operatorname{Sing}(B_f)}$.

Let $(a, b) \in S$. Then $\sharp f^{-1}(a, b) = 1$ because $\sharp S = \sharp f^{-1}(S)$. So, $\sharp f_1^{-1}(a) = 1$ and hence $a \in B_{f_1}$. Similarly, $b \in B_{f_2}$. Therefore, $+_{(a,b)} \subseteq B_f$. □

Notation 2.18. Let X be a sextic del Pezzo surface. Denote by $\hexagon_X$ the union of all the six (-1)-curves on X.

Lemma 2.19. *Let X be a sextic del Pezzo surface. Let A be a reduced ample divisor such that $\operatorname{Supp} A \subseteq \hexagon_X$. Then $A = \hexagon_X$.*

Proof. Write $\hexagon_X = \bigcup_{i=1}^{6}\ell_i$ such that $\ell_i\cap\ell_j \neq \emptyset$ if and only if $i - j \equiv \pm 1 \mod 6$. Since A is ample, $\operatorname{Supp} A$ is connected. Then we may write $A = \sum_{i=1}^{m}\ell_i$. Suppose $m < 6$. Then $A\cdot\ell_1 = 0$, a contradiction. □

Lemma 2.20. *Let $f : X \to X$ be a surjective endomorphism of a sextic del Pezzo surface X. Then $f^{\pm}(\hexagon_X) = \hexagon_X$. Suppose further f is nonisomorphic. Then $B_f = \hexagon_X$.*

Proof. Note that $\hexagon_X = \bigcup_{i=1}^{6}\ell_i$ is the union of all negative curves. The first argument is then obvious (cf. [39, Lemma 4.3], [44, Proposition 11]).

Write $f^*\ell_i = q\ell_j$. Then $q = (\deg f)^{\frac{1}{2}}$ by the projection formula. So, $f^*(\hexagon_X) = q\hexagon_X$. Since $\deg f > 1$, we have $q > 1$ and hence $\hexagon_X \subseteq B_f$. Note that $K_X + \hexagon_X \sim 0$. By the ramification divisor formula, we have $K_X + \hexagon_X = f^*(K_X + \hexagon_X)$ and hence $\hexagon_X = B_f$. □

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2.4 | Extremal contractions of smooth projective threefolds

We adapt Mori's well-known results [35] on smooth projective threefolds to our specific case for the convenience of the reader.

Theorem 2.21. *Let X be a smooth rationally connected projective threefold of Picard number two. Let $\pi : X \to Y$ be an extremal K_X-negative birational contraction with the exceptional locus E. Let C_π be a π-contracted curve whose anti-canonical degree $-K_X \cdot C_\pi$ is minimal. Then one of the following types occur.*

- *[E1] The π is the blowup of a smooth Fano threefold Y along a smooth curve $\ell \subseteq Y$ and E is a $\mathbb{P}^1$-bundle over ℓ such that $K_X = \pi^* K_Y + E$ and $K_X \cdot C_\pi = E \cdot C_\pi = -1$.*
- *[E2] The π is the blowup of a smooth Fano threefold Y along a point such that $K_X = \pi^* K_Y + 2E$ and $E \cong \mathbb{P}^2$ with $K_X \cdot C_\pi = 2E \cdot C_\pi = -2$.*
- *[E3] The π is the blowup of a factorial terminal Fano threefold Y along a nodal point z such that $K_X = \pi^* K_Y + E$ and $E \cong \mathbb{P}^1 \times \mathbb{P}^1$ with $K_X \cdot C_\pi = E \cdot C_\pi = -1$.*
- *[E4] The π is the blowup of a factorial terminal Fano threefold Y along a cuspidal point z such that $K_X = \pi^* K_Y + E$ and E is a singular quadric surface with $K_X \cdot C_\pi = E \cdot C_\pi = -1$.*
- *[E5] The π is the blowup of a 2-factorial terminal Fano threefold Y along a $\frac{1}{2}(1,1,1)$-singularity z such that $K_X = \pi^* K_Y + \frac{1}{2}E$ and $E \cong \mathbb{P}^2$ with $K_X \cdot C_\pi = \frac{1}{2}E \cdot C_\pi = -1$.*

Moreover, for the types E1 ~ E4, the anti-canonical divisor $-K_Y$ is base-point-free ([30, Proposition 4.1]; cf. [25]).

Remark 2.22. It is worth noting that, as proved in [30, Theorems 1.7 and 1.8], if a smooth rationally connected projective threefold X of Picard number two admits a divisorial contraction of type E3 or E4, then X itself is weak Fano. In our paper, we try to give a self-contained proof from the dynamical viewpoints and we shall only use Theorems 2.21 and 2.23.

Theorem 2.23 [35, Theorem 3.5]. *Let X be a smooth rationally connected projective threefold of Picard number two. Let $\pi : X \to Y$ be an extremal Fano contraction. Then one of the following types occur.*

- *[C1] π is a conic bundle over $Y \cong \mathbb{P}^2$ with the discriminant locus $\Delta \neq \emptyset$.*
- *[C2] there exists a rank two locally free sheaf $\mathcal{E}$ on $Y \cong \mathbb{P}^2$ such that π is the natural projection $X = \mathbb{P}_Y(\mathcal{E}) \to Y$ (cf. [11, Proposition and Corollary 1]).*
- *[D1] π is a del Pezzo fibration over a smooth curve Y whose general fiber is a del Pezzo surface of degree $\leqslant 6$.*
- *[D2] π is a quadric surface fibration over a smooth curve Y (cf. Definition 5.6).*
- *[D3] there exists a rank three locally free sheaf $\mathcal{E}$ on $Y \cong \mathbb{P}^1$ such that π is the projection $X = \mathbb{P}_Y(\mathcal{E}) \to Y$ (cf. [11, Proposition and Corollary 1]).*

3 | FRAMEWORK FOR SETTING UP A PROOF

Let X be a smooth Fano type threefold with $\rho(X) = 2$. Say (X, Δ) is log Fano. With the cone theorem [28, Theorem 3.7] applied to (X, Δ), the two extremal rays of $\overline{\mathrm{NE}}(X)$ are both contractible.

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Therefore, we can introduce the following notation. It will help us better elaborate on our proof in the subsequent sections.

Notation 3.1. Denote by $\mathcal{S}$ the collection of $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z)$ satisfying:

(1) X is a smooth projective threefold of Fano type with $\rho(X) = 2$;
(2) π is the contraction of some K_X-negative extremal ray with $\dim Y$ minimal among all the K_X-negative contractions;
(3) τ is the contraction of another (not necessarily K_X-negative) extremal ray; and
(4) f is an int-amplified endomorphism of X with π and τ being f-equivariant (see [38, Theorem 1.1]).

Denote by

$$\mathcal{S}_{\text{Fano}} := \{(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S} \mid \pi \text{ is Fano}\}$$

Denote by

$$\mathcal{S}_{\text{bir}} := \{(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S} \mid \pi \text{ and } \tau \text{ are birational}\}$$

Lemma 3.2. $\mathcal{S} = \mathcal{S}_{\text{Fano}} \cup \mathcal{S}_{\text{bir}}$.

Proof. Let $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S} \setminus \mathcal{S}_{\text{Fano}}$. Then π is not Fano and hence a birational morphism.

Suppose the contrary that τ is not birational. Then $\dim X > \dim Z$. Let Δ be an effective $\mathbb{Q}$-Cartier Weil $\mathbb{Q}$-divisor such that (X, Δ) is klt and $-(K_X + \Delta)$ is ample. Note that a general fiber X_z is not contained in $\operatorname{Supp}\Delta$. Then there exists some curve $C \subseteq X_z$ such that $C \nsubseteq \operatorname{Supp}\Delta$. In particular, $\Delta \cdot C \geqslant 0$. Then $K_X \cdot C \leqslant (K_X + \Delta) \cdot C < 0$ and hence τ is a contraction of some K_X-negative extremal ray with $\dim Z < \dim Y$. This is a contradiction to the minimality of $\dim Y$. □

Remark 3.3. By Lemma 3.2 and Theorem 2.4, it suffices to show Theorem 1.3 for elements in $\mathcal{S}_{\text{Fano}}$ and $\mathcal{S}_{\text{bir}}$, respectively.

Lemma 3.4. *Let $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}_{\text{bir}}$. Then π is not of type E2 in Theorem 2.21.*

Proof. Suppose the contrary that π is of type E2. It follows from Theorem 2.21 that Y is a smooth Fano threefold of Picard number 1 and π is the blowup of Y along a point. By [33, Theorem 1.10], after iteration, f descends to a nonisomorphic surjective endomorphism on Y and hence Y is isomorphic to $\mathbb{P}^3$, see [2] or [20]. But then, X (as the blowup of $\mathbb{P}^3$ at a point) admits a $\mathbb{P}^1$ bundle over $\mathbb{P}^2$, a contradiction to the choice of τ. □

Theorem 3.5 [43, Theorem 5.1]. *Let $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}$ with X being Fano. Then X is isomorphic to either $\mathbb{P}_{\mathbb{P}^2}(\mathcal{O}_{\mathbb{P}^2} \oplus \mathcal{O}_{\mathbb{P}^2}(e))$ with $e \leqslant 0$ or a blowup of $\mathbb{P}^3$ along a line. In particular, we have $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}_{\text{Fano}}$.*

Proof. We only need to notice that the X is a $\mathbb{P}^2$-bundle over $\mathbb{P}^1$ when it is the blowup of $\mathbb{P}^3$ along a line. □

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4 | FANO CONTRACTION TO A SURFACE, PROOF OF THEOREM 4.1

In this section, we treat $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}_{\mathrm{Fano}}$ (cf. Notation 3.1) where $\dim Y = 2$. We aim to show:

Theorem 4.1. *Let $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}_{\mathrm{Fano}}$ with $\dim Y = 2$. Then $X \cong \mathbb{P}_Y(\mathcal{O}_Y \oplus \mathcal{O}_Y(e))$ for some $e < 0$ where $Y \cong \mathbb{P}^2$. In particular, X is toric.*

We first show that π is smooth.

Lemma 4.2. *Under the assumption of Theorem 4.1, the discriminant locus Δ of π is empty.*

Proof. It follows from Theorem 2.23 that π is a conic bundle over $Y \cong \mathbb{P}^2$. Suppose the contrary that $\Delta \neq \emptyset$.

Note that Δ is a divisor. The fibers X_y with $y \in \Delta$ satisfy the following: If y is a smooth point of Δ, then X_y is a reducible conic in $\mathbb{P}^2$. If y is a double point of Δ, then X_y is isomorphic to a double line in $\mathbb{P}^2$ (cf. [4, Proposition 1.2]).

By [10, Lemmas 7.2 and 7.4], we have $g^{-1}(\Delta) = \Delta$. Write $\Delta = \bigcup_{i=1}^{m} C_i$ where C_i are irreducible curves. By [19, Theorem], each C_i is a line. Let $P_i := \pi^{-1}(C_i)$. As $\pi \circ f = g \circ \pi$, we see that P_i is f^{-1}-periodic. By the cone theorem [28, Theorem 3.7], $P_i = \pi^* C_i$ is a prime divisor. Note that a general fiber of $\pi|_{P_i} : P_i \to C_i \cong \mathbb{P}^1$ has two irreducible components. Then $\sharp B_{\pi|_{P_i}} \geqslant 2$ by the Hurwitz formula (cf. Definition 2.5). Note that for any $y \in B_{\pi|_{P_i}}$, the fiber X_y is a double line; in particular, $\sharp C_i \cap \mathrm{Sing}(\Delta) \geqslant 2$ for each $1 \leqslant i \leqslant m$ and hence $m \geqslant 3$. We only need $m \geqslant 2$. Note that $P_2|_{P_1} = (\pi|_{P_1})^*(C_2|_{C_1})$ is a double line. So, we get a contradiction to Lemma 2.9. □

Proof of Theorem 4.1. By Lemma 4.2, π is smooth $\mathbb{P}^1$-fibration. By Theorem 2.23, there exists a rank two locally free sheaf $\mathcal{E}$ on Y such that $X \cong \mathbb{P}_Y(\mathcal{E})$. The splitting of $\mathcal{E}$ is then guaranteed by [1, Proposition 3]. Finally, note that $X \ncong \mathbb{P}^2 \times \mathbb{P}^1$ by Notation 3.1. So, $e < 0$. □

The following corollary will be used in the proof of Claim 8.15.

Corollary 4.3. *The second contraction of X in Theorem 4.1 is never small.*

Proof. Let $E \cong Y$ be a section determined by the projection

$$\mathcal{O}_Y \oplus \mathcal{O}_Y(e) \to \mathcal{O}_Y(e).$$

Then $\mathcal{O}_X(E)|_E \cong \mathcal{O}_{\mathbb{P}^2}(e)$. So, τ is the E-negative contraction with E being the exceptional divisor. □

5 | DISCRIMINANT LOCI OF QUADRIC SURFACE FIBRATIONS

In this section, we describe the discriminant locus of a quadric surface fibration with the help of non-$\mathbb{Q}$-factorial loci. We aim to show Theorem 5.11 prepared for Lemma 6.4.

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5.1 | Non-$\mathbb{Q}$-factorial loci

We recall several facts on the non-$\mathbb{Q}$-factorial locus treated at the end of [34, section 6]. A local ring $(R, \mathfrak{m})$ is $\mathbb{Q}$-*factorial* if for any prime ideal $\mathfrak{p}$ of height one, $\mathfrak{p} = \sqrt{(f)}$ for some $f \in \mathfrak{p}$.

Definition 5.1 [34, Lemma 6.20]. Let X be a normal variety. Denote by $\hat{\mathcal{O}}_{X,x}$ the completion of the stalk. Denote by

$$\mathrm{NQF}(X) = \{x \in X \mid \hat{\mathcal{O}}_{X,x} \text{ is not } \mathbb{Q}\text{-factorial}\}$$

the *non-$\mathbb{Q}$-factorial locus* of X. Here we only consider closed points. Note that $\mathrm{NQF}(X) \subseteq \mathrm{Sing}(X)$.

In general, it is not clear whether $\mathrm{NQF}(X)$ is a closed subset of X or not. The reason we work on the completion ring is due to the following functorial result.

Lemma 5.2 [34, Lemma 6.20]. *Let $f : X \to Y$ be a finite surjective morphism of normal varieties. Then $f^{-1}(\mathrm{NQF}(Y)) \subseteq \mathrm{NQF}(X)$.*

From the following lemma, we know that the non-$\mathbb{Q}$-factorial locus being empty implies the usual $\mathbb{Q}$-factoriality of a variety.

Lemma 5.3 [34, Lemma 6.21]. *Let X be a normal variety and $x \in X$ a closed point. Then $\hat{\mathcal{O}}_{X,x}$ being $\mathbb{Q}$-factorial implies that $\mathcal{O}_{X,x}$ is $\mathbb{Q}$-factorial; the variety X is $\mathbb{Q}$-factorial if $\mathcal{O}_{X,x}$ is $\mathbb{Q}$-factorial for every $x \in X$.*

5.2 | Normal morphisms

Definition 5.4. Let $\pi : X \to Y$ be a morphism of varieties. Let $x \in X$, a not necessarily closed point, and $y = \pi(x)$. We say that π is *normal* at x if π is flat at x, and the scheme-theoretical fiber X_y is geometrically normal at x over $k(y)$. We say that π is a *normal morphism* if π is normal at every point of X.

Suppose π is flat. By [18, Theorem 12.1.6], the set $\{y \in Y \mid X_y \text{ is normal}\}$ is open. In particular, a flat morphism π is normal if and only if X_y is normal for every closed point $y \in Y$.

Proposition 5.5 [17, Proposition 6.8.3]. *Normal morphisms are stable under base change and composition.*

5.3 | Quadric surface fibrations

Definition 5.6. Let $\pi : X \to Y$ be a projective morphism between varieties. We say π is a *quadric surface fibration* if the following conditions hold.

(1) π is flat.
(2) Each fiber of π is isomorphic to a (possibly singular) quadric surface in $\mathbb{P}^3$.
(3) A general fiber of π is smooth.

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The *discriminant* Δ of π is the set of points over which π is not smooth.

Since flat morphisms are stable under base change, we also have

Proposition 5.7. *Quadric surface fibrations are stable under base change.*

Proposition 5.8. *Let $\pi : X \to Y$ be a quadric surface fibration with Y being normal. Then X is normal.*

Proof. Note that a singular quadric surface in $\mathbb{P}^3$ is normal. Then π is a normal morphism. By Proposition 5.5, we see that X is normal. □

Lemma 5.9. *Let $\pi : X \to Y$ be a quadric surface fibration. Suppose the restriction $\mathrm{Pic}_{\mathbb{Q}}(X) \to \mathrm{Pic}_{\mathbb{Q}}(X_y)$ is surjective for a general $y \in Y$. Then π is smooth.*

Proof. Let ℓ be a line in $X_y \cong \mathbb{P}^1 \times \mathbb{P}^1$ with y general. By the assumption, there exists a $\mathbb{Q}$-Cartier divisor D on X such that $D|_{X_y} \sim_{\mathbb{Q}} \ell$. Let A be a π-ample divisor on X. Then

$$0 < D \cdot A \cdot X_y = D \cdot A \cdot X_{y'},$$

$$0 = D^2 \cdot X_y = D^2 \cdot X_{y'}$$

for any $y' \in Y$.

Suppose the contrary that π is not smooth. Then $X_{y'}$ is a singular quadric surface in $\mathbb{P}^3$ for some $y' \in Y$. Note that $\rho(X_{y'}) = 1$. Then we have $D|_{X_{y'}} \equiv 0$ by the vanishing $D^2 \cdot X_{y'} = 0$. This contradicts $D|_{X_{y'}} \cdot A|_{X_{y'}} > 0$. □

Now we focus on the situation when the base is a curve.

Lemma 5.10. *Let $\pi : X \to Y$ be a quadric surface fibration where Y is smooth of dimension one. Suppose the restriction $\mathrm{Pic}_{\mathbb{Q}}(X_{\mathrm{reg}}) \to \mathrm{Pic}_{\mathbb{Q}}(X_y)$ is surjective for general y. Then $\mathrm{NQF}(X) \cap X_y \neq \emptyset$ if X_y is singular.*

Proof. By Proposition 5.8, X is normal. Suppose the contrary that $\mathrm{NQF}(X) \cap X_{y'} = \emptyset$ for some singular fiber $X_{y'}$. Since $\dim Y = 1$, there are only finitely many singular fibers. After shrinking Y, we may assume π has exactly the singular fiber $X_{y'}$. Then $\mathrm{NQF}(X) = \emptyset$ and hence X is $\mathbb{Q}$-factorial by Lemma 5.3. In particular, the restriction $\mathrm{Pic}_{\mathbb{Q}}(X) \to \mathrm{Pic}_{\mathbb{Q}}(X_{\mathrm{reg}})$ is surjective. By the assumption, we see that $\mathrm{Pic}_{\mathbb{Q}}(X) \to \mathrm{Pic}_{\mathbb{Q}}(X_y)$ is surjective if X_y smooth. By Lemma 5.9, π is smooth, a contradiction. □

Theorem 5.11. *Let $\pi : X \to Y$ be a quadric surface fibration over a smooth projective curve Y. Suppose X admits a surjective endomorphism f such that π is f-equivariant with the induced endomorphism g on Y. Suppose further the restriction $\mathrm{Pic}_{\mathbb{Q}}(X_{\mathrm{reg}}) \to \mathrm{Pic}_{\mathbb{Q}}(X_y)$ is surjective for general y. Then the discriminant locus Δ of π is g^{-1}-invariant.*

Proof. By Lemma 5.2, we have $f^{-1}(\mathrm{NQF}(X)) \subseteq \mathrm{NQF}(X)$. Let $S = \overline{\mathrm{NQF}(X)} \subseteq \mathrm{Sing}(X)$. Note that X is normal by Proposition 5.8. Then $f^{-1}(S) \subseteq S$ by [10, Lemma 7.2] and hence $f^{-1}(S) = S$. By Lemma 5.10, we have $\pi(S) = \Delta$. Then we are done by [10, Lemma 7.5]. □

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6 | FANO CONTRACTION TO A CURVE, PROOF OF THEOREM 6.1

In this section, we deal with $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}_{\mathrm{Fano}}$ (cf. Notation 3.1) where $\dim Y = 1$. We aim to show:

Theorem 6.1. *Let $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}_{\mathrm{Fano}}$ with $\dim Y = 1$. Then π is a $\mathbb{P}^2$-bundle over $Y \cong \mathbb{P}^1$; in particular, X is toric.*

Proof. By [33, Theorem 1.10], f descends to an int-amplified endomorphism g on Y. By [12, Theorem 5.1], the set of g-periodic points $\mathrm{Per}(g)$ is Zariski dense in Y. Let $U \subseteq Y$ be a nonempty Zariski open subset such that X_y is smooth for $y \in U$. For $y \in \mathrm{Per}(g) \cap U \neq \emptyset$, there exists $s_y > 0$ such that $g^{s_y}(y) = y$ and hence $f^{s_y}(X_y) = X_y$. By [33, Lemma 2.2], $f^{s_y}|_{X_y}$ is still int-amplified. Then X_y is a smooth toric del Pezzo surface by [44, Theorem 3]. By further applying Theorem 2.23, the general fiber of π belongs to one of the three cases: $\mathbb{P}^2$, a smooth quadric surface $\mathbb{P}^1 \times \mathbb{P}^1$, or a del Pezzo surface of degree 6. The latter two cases are excluded in Theorems 6.3 and 6.6. Therefore, X is a smooth $\mathbb{P}^2$-bundle over $\mathbb{P}^1$; in particular, by the triviality of the Brauer group of $\mathbb{P}^1$ as well as Grothendieck's splitting theorem, our X is a splitting $\mathbb{P}^2$-bundle over $\mathbb{P}^1$ and hence toric by [9, (7.3.1), p. 337]. □

The following corollary will be used in the proof of Claim 8.15.

Corollary 6.2. *Suppose further in Theorem 6.1 that $-K_X$ is nef and the second contraction τ is small. Then $\kappa(X, D) \geqslant 1$ for any nonzero effective divisor D on X.*

Proof. Let $X \cong \mathbb{P}_{\mathbb{P}^1}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(a) \oplus \mathcal{O}_{\mathbb{P}^1}(b))$ with $0 \geqslant a \geqslant b$ in Theorem 6.1. Since τ is small, we have $b < 0$. Let E be the divisor determined by the projection

$$\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(a) \oplus \mathcal{O}_{\mathbb{P}^1}(b) \to \mathcal{O}_{\mathbb{P}^1}(a) \oplus \mathcal{O}_{\mathbb{P}^1}(b).$$

Let C be a section determined by the projection

$$\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(a) \oplus \mathcal{O}_{\mathbb{P}^1}(b) \to \mathcal{O}_{\mathbb{P}^1}(b).$$

Then $E \cdot C = b$. By the relative Euler sequence, we have

$$-K_X \sim 3E - (a + b - 2)F$$

where F is a fiber of π.

Since $-K_X$ is nef, we have $-K_X \cdot C = 2b - a + 2 \geqslant 0$. If $a = b < 0$, then $E \cong \mathbb{P}^1 \times \mathbb{P}^1$ and the E-negative curves cover E, a contradiction because τ is small. So, $a > b$ and hence $(a, b) = (0, -1)$. Note that

$$h^0(X, nE) = h^0(\mathbb{P}^1, \mathrm{Sym}^n(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1))) = n + 1.$$

Therefore, we have $\kappa(X, E) = 1$. Note that $\mathrm{PE}(X)$ is generated by E and F where F is the fiber divisor of π with $\kappa(X, F) = 1$. Consequently, for any nonzero effective divisor D, its Iitaka dimension satisfies $\kappa(X, D) \geqslant 1$. □

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6.1 | Quadric surface fibration case, Proof of Theorem 6.3

Theorem 6.3. *Under the assumption of Theorem 6.1, the general fiber of π is not isomorphic to a smooth quadric surface.*

We prepare the following lemma first.

Lemma 6.4. *Under the assumption of Theorem 6.1 with the general fiber of π isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$, after suitable iteration of f, we have the following.*

(1) *The second contraction $\tau : X \to Z$ is birational and X is not Fano.*
(2) *There exists an f^{-1}-invariant prime divisor P dominating Y.*
(3) *For any y not in the discriminant locus Δ of π, the intersection set $X_y \cap P = +_{\mathrm{Sing}(X_y \cap P)}$ (cf. Notation 2.14) and the restriction $P|_{X_y}$ is a divisor of type (p, p) in $\mathbb{P}^1 \times \mathbb{P}^1$ with $p = 1$ or 2.*
(4) *Let $\nu : P^n \to P$ be the normalization and $P^n \xrightarrow{\phi} \widetilde{Y} \xrightarrow{\sigma} Y$ the Stein factorization of $\pi|_P \circ \nu$. Then $\sharp B_\sigma \geqslant 2$ (cf. Definition 2.5).*
(5) *The discriminant locus Δ is g^{-1}-invariant, where $g : Y \to Y$ is induced by f.*

Proof. By Theorem 2.23, every fiber of π is an irreducible and reduced quadric in $\mathbb{P}^3$. So, π is a quadric surface fibration as we defined in Definition 5.6.

Let $y \notin \Delta$. Then $X_y \cong \mathbb{P}^1 \times \mathbb{P}^1$. By the adjunction formula, the divisor $K_X|_{X_y} = K_{X_y}$ is of type $(-2, -2)$ on $\mathbb{P}^1 \times \mathbb{P}^1$. Let D be any Cartier divisor on X. Since $\rho(X) = \rho(Y) + 1$, we have $D - tK_X \in \pi^* \mathrm{N}^1(Y)$ for some rational number t. In particular, $D|_{X_y}$ is of type $(-2t, -2t)$ on $\mathbb{P}^1 \times \mathbb{P}^1$.

(1) Suppose the contrary $\dim X > \dim Z$. Then τ is also a K_X-negative contraction of an extremal ray. By Kleiman's ampleness criterion, $-K_X$ is ample. By Theorem 3.5, it is only possible that $X \cong \mathbb{P}_{\mathbb{P}^2}(\mathcal{O}_{\mathbb{P}^2} \oplus \mathcal{O}_{\mathbb{P}^2}(e))$ for some $e \leqslant 0$. However, $\mathbb{P}_{\mathbb{P}^2}(\mathcal{O}_{\mathbb{P}^2} \oplus \mathcal{O}_{\mathbb{P}^2}(e))$ admits a section isomorphic to $\mathbb{P}^2$. Note that there is no fiber of π isomorphic to $\mathbb{P}^2$. Then we obtain $\mathbb{P}^2$ dominating $Y \cong \mathbb{P}^1$, a contradiction.
(2) By (1), let E be the τ-exceptional locus. Then $f^{-1}(E) = E$ and E dominates Y. If $\dim E = 2$, then E is a prime divisor by [28, Proposition 2.5] and we simply let $P = E$. So, we may assume $\dim E = 1$ and let $C \subseteq E$ be an irreducible curve. After iteration, we may assume $f^{-1}(C) = C$. Define

$$\Sigma := \{y \in Y \mid \sharp(\pi|_C)^{-1}(y) < \deg \pi|_C\}.$$

Since Y is a smooth curve, we see that Σ is a finite set.

We claim that $g(\Delta) \subseteq \Delta$ and $g(\Sigma) \subseteq \Sigma$. For any $y', y \in Y$ with $g(y') = y$, we have that $f|_{X_{y'}} : X_{y'} \to X_y$ is surjective by [10, Lemma 7.3]. Note that $\rho(X_y) = 2$ if $y \notin \Delta$ and $\rho(X_y) = 1$ if $y \in \Delta$. So, $g(\Delta) \subseteq \Delta$. Note that $f^{-1}(C) = C$. Then $f|_{(\pi|_C)^{-1}(y')} : (\pi|_C)^{-1}(y') \to (\pi|_C)^{-1}(y)$ is also surjective. In particular, $\sharp(\pi|_C)^{-1}(y') \geqslant \sharp(\pi|_C)^{-1}(y)$ and hence $g(\Sigma) \subseteq \Sigma$.

For a subset $S \subseteq \mathbb{P}^1 \times \mathbb{P}^1$, recall Notation 2.14 that $+_S$ is the union of crosses with anchored crossing points in S. Construct a closed subset as follows

$$Q := \overline{\bigcup_{y \notin \Delta \cup \Sigma} +_{X_y \cap C}}.$$

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Note that Q contains C. Let $y \notin \Delta \cup \Sigma$ and $y' \in g^{-1}(y)$. Then $y' \notin \Delta \cup \Sigma$. By Lemma 2.16, we have

$$(f|_{X_{y'}})^{-1}(+_{X_y \cap C}) = +_{X_{y'} \cap C}$$

and hence

$$f^{-1}\left(\bigcup_{y \notin \Delta \cup \Sigma} +_{X_y \cap C}\right) \subseteq \bigcup_{y \notin \Delta \cup \Sigma} +_{X_y \cap C}.$$

By lemma [10, Lemma 7.2], we have $f^{-1}(Q) \subseteq Q$ and hence $f^{-1}(Q) = Q$.

Let $U \subseteq Y \backslash (\Delta \cup \Sigma)$ be a nonempty open subset of Y such that $X_y \nsubseteq B_f$ and $B_f|_{X_y}$ is reduced for any $y \in U$. Let $y \in U$ and $y' \in g^{-1}(y)$. Then we have

$$B_{f|_{X_{y'}}} = B_f|_{X_y}.$$

Let $A = \tau^* A_Z$ where A_Z is an ample divisor on Z. As $\rho(Z) = 1$, we have $(f|_Z)^* A_Z \equiv q A_Z$ for some integer $q > 1$. Then $f^* A \equiv qA$ and A is π-ample. Note that $A|_{X_y}$ and $A|_{X_{y'}}$ are of the same type (d, d) on $\mathbb{P}^1 \times \mathbb{P}^1$ for some $d > 0$. Moreover, we have $(f|_{X_{y'}})^*(A|_{X_y}) \equiv qA|_{X_{y'}}$. In particular, $f|_{X_{y'}}$ induces a polarized endomorphism after we identify $X_{y'}$ with X_y via an arbitrary isomorphism. Note that

$$\sharp X_{y'} \cap C = \sharp (f|_{X_{y'}})^{-1}(X_y \cap C) = \sharp X_y \cap C = \deg \pi|_C.$$

By Lemma 2.17, we have

$$+_{X_y \cap C} \subseteq B_{f|_{X_{y'}}} \subseteq B_f$$

and hence

$$Q \cap \pi^{-1}(U) \subseteq B_f.$$

By the construction, some irreducible component P of Q is a prime divisor dominating Y. After iteration, $f^{-1}(P) = P$.

(3) Let U be the open set in (2). Note that $P \cap X_y \subseteq B_{f|_{X_{y'}}} = +_{\mathrm{Sing}(B_{f|_{X_{y'}}})}$ for any $y \in U$ with $y' \in g^{-1}(U)$ by Lemma 2.17. Note that $P|_{X_y}$ is ample. By Lemma 2.15, we have $X_y \cap P = +_{\mathrm{Sing}(X_y \cap P)}$. Now let $y \in Y \backslash (\Delta \cup U)$. If $g^{-1}(y) = y$ after iteration, then $P|_{X_y} = X_y \cap P \subseteq B_{f|_{X_y}}$ is a reduced ample divisor (see Lemma 2.9), and then the same argument holds. If y is not g^{-1}-periodic, then there exists some $y' \in U$ with $g^s(y') = y$ for some $s > 0$. Note that $f^s(X_{y'} \cap P) = X_y \cap P$. Then $X_y \cap P = +_{\mathrm{Sing}(X_y \cap P)}$ by Lemma 2.16.

By Lemma 2.8, $-(K_X + P)$ is pseudo-effective. Let $y \notin \Delta$. By the adjunction formula, it follows that $-(K_{X_y} + P|_{X_y})$ is pseudo-effective. So, $P|_{X_y}$ is a divisor of type (p, p) with $p = 1$ or 2.

(4) By (3), the general fiber of $\pi|_P : P \to Y$ has $2p$ irreducible components. So, $\deg \sigma = 2p \geqslant 2$. Since $Y \cong \mathbb{P}^1$, we have $\sharp B_\sigma \geqslant 2$ by the simply connectedness of $\mathbb{P}^1$ and $\mathbb{A}^1$.

(5) Let $\widetilde{X} := X \times_Y \widetilde{Y}$ and denote by $\widetilde{\pi} : \widetilde{X} \to \widetilde{Y}$ and $\widetilde{\sigma} : \widetilde{X} \to X$ the induced morphisms. Then $\widetilde{\pi}$ is again a quadric surface fibration and $\widetilde{X}$ is normal by Proposition 5.8. By the diagonal

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embedding $\theta : \widetilde{Y} \hookrightarrow \widetilde{Y} \times_Y \widetilde{Y}$, the irreducible component $\theta(\widetilde{Y})$ is a section over $\widetilde{Y}$. Let $\widetilde{P}_1$ be the irreducible component of $P \times_Y \widetilde{Y}$ dominating $\theta(\widetilde{Y})$ via the dominant rational map $P \times_Y \widetilde{Y} \dashrightarrow \widetilde{Y} \times_Y \widetilde{Y}$. Note that the dominant rational map $P \dashrightarrow \widetilde{Y}$ has irreducible and reduced general fibers. Then $\widetilde{P}_1 \cap \widetilde{X}_{\widetilde{y}}$ is a line for general $\widetilde{y} \in \widetilde{Y}$. In particular, the restriction map $\mathrm{Pic}_{\mathbb{Q}}(\widetilde{X}_{\mathrm{reg}}) \to \mathrm{Pic}_{\mathbb{Q}}(\widetilde{X}_{\widetilde{y}})$ is surjective because $\mathrm{Pic}_{\mathbb{Q}}(\widetilde{X}_{\widetilde{y}}) = \langle \widetilde{P}_1|_{\widetilde{X}_{\widetilde{y}}}, K_{\widetilde{X}}|_{\widetilde{X}_{\widetilde{y}}} \rangle$.

Note that $f|_P$ lifts to a surjective endomorphism of P^n. By [10, Lemma 5.2], there is a surjective endomorphism $\widetilde{g} : \widetilde{Y} \to \widetilde{Y}$ such that $\sigma \circ \widetilde{g} = g \circ \sigma$. So, f lifts to a surjective endomorphism $\widetilde{f}$ of $\widetilde{X}$. Denote by $\widetilde{\Delta}$ the discriminant locus of $\widetilde{\pi}$. By Theorem 5.11, we have $\widetilde{g}^{-1}(\widetilde{\Delta}) = \widetilde{\Delta}$. Note that $\widetilde{\Delta} = \sigma^{-1}(\Delta)$. Then $g^{-1}(\Delta) = \Delta$.

Lemma 6.4 is proved. □

Remark 6.5. Lemma 6.4(3) does not imply that $P|_{X_y}$ is a reduced divisor for $y \notin \Delta$.

Proof of Theorem 6.3. We adopt Lemma 6.4 and use its notation there.

We claim that $(X, P + X_{y_0})$ is lc for any $y_0 \in B_{\pi|_P} \neq \emptyset$. Once the claim is proved, we obtain a contradiction to Corollary 2.7.

Case: $P|_{X_y}$ is of type (2,2) for $y \notin \Delta$. Then $(K_X + P)|_{X_y} \sim 0$. Since π is a K_X-negative contraction of relative Picard number one, by the cone theorem [28, Theorem 3.7], we obtain that $K_X + P \sim \pi^* L$ for some line bundle L on Y. By Corollary 2.12, it follows that $(X, P + X_y)$ is lc for any $y \in Y$.

Case: $P|_{X_y}$ is of type (1,1) for $y \notin \Delta$. Then $P|_{X_y}$ is a reduced divisor supported by a single cross $+_{o_y}$ with o_y being the point of the crossing. Note that after normalization ν, each cross $P|_{X_y}$ becomes two disjoint lines. In particular, we have $B_\sigma \subseteq \Delta$. Note also that $g^{-1}(\Delta) = \Delta$ and $B_\sigma \neq \emptyset$. Let $y_0 \in B_{\pi|_P} = B_\sigma \neq \emptyset$. Then $f^{-1}(X_{y_0}) = X_{y_0}$ after iteration. So, $(X, P + X_{y_0})$ is lc by Lemma 2.8. □

6.2 | Sextic del Pezzo fibration case, Proof of Theorem 6.6

Theorem 6.6. *Under the assumption of Theorem 6.1, the general fiber of π is not isomorphic to a sextic del Pezzo surface.*

We prepare the following lemma first like Lemma 6.4, but there are some differences. We are not able to show the total invariance of the discriminant locus. However, we can nevertheless show the total invariance of the horizontal part of the branch divisor.

Lemma 6.7. *Under the assumption of Theorem 6.1 with the general fiber of π isomorphic to a sextic del Pezzo surface, we have the following.*

(1) *Let P be the π-horizontal part of B_f. Then P is an f^{-1}-invariant prime divisor.*
(2) *For any y not in the discriminant locus Δ of π, the divisor $P|_{X_y} = \hexagon_{X_y}$ (cf. Notation 2.18).*
(3) *Let $\nu : P^n \to P$ be the normalization and $P^n \xrightarrow{\phi} \widetilde{Y} \xrightarrow{\sigma} Y$ the Stein factorization of $\pi|_P \circ \nu$. Then $\sharp B_\sigma \geqslant 2$.*

Proof. Let S be a sextic del Pezzo surface. Recall Notation 2.18 that $\hexagon_S$ is the union of all the six (-1) curves in S. Then $-K_S \sim \hexagon_S$. Note that X_y is isomorphic to S if and only if $y \notin \Delta$. Let $y \notin \Delta$.

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By the adjunction formula, we have $-K_X|_{X_y} = -K_{X_y} \sim \mathcal{O}_{X_y}$. Let D be any Cartier divisor on X. Since $\rho(X) = \rho(Y) + 1$, we have $D - tK_X \in \pi^* \mathrm{N}^1(Y)$ for some rational number t. In particular, $D|_{X_y} \equiv -t\mathcal{O}_{X_y}$.

Denote by $g : Y \to Y$ the induced endomorphism after iteration.

(1) Let $U \subseteq Y\backslash(\Delta \cup g(\Delta))$ be a nonempty open subset of Y such that $X_y \nsubseteq B_f$ and $B_f|_{X_y}$ is reduced for any $y \in U$. Let $y \in U$ and $y' \in g^{-1}(y)$. The restriction $f|_{X_{y'}} : X_{y'} \to X_y$ is surjective by [10, Lemma 7.3]. Since f is int-amplified, there exists a π-ample divisor A such that $f^*A \equiv qA$ for some integer $q > 1$. Moreover, we have $(f|_{X_{y'}})^*(A|_{X_y}) \equiv qA|_{X_{y'}}$. In particular, $f|_{X_{y'}}$ induces a polarized endomorphism after we identify $X_{y'}$ with X_y via an arbitrary isomorphism. By Lemma 2.20, we have

$$(f|_{X_{y'}})^{-1}(\mathcal{O}_{X_y}) = \mathcal{O}_{X_{y'}}$$

and

$$\mathcal{O}_{X_y} = B_{f|_{X_{y'}}} = P \cap X_y$$

where P is the π-horizontal part of B_f. Then

$$f^{-1}(P \cap \pi^{-1}(U)) \cap \pi^{-1}(U) \subseteq P$$

and hence $f^{-1}(P) = P$, see [10, Lemma 7.2].

Let P_1 be an irreducible component of P. Note that P_1 dominates Y. Then P_1 is π-ample. Let $y \in U$. Then $P_1|_{X_y} \subseteq \mathcal{O}_{X_y}$ is a reduced ample divisor. By Lemma 2.19, we have $P_1|_{X_y} = \mathcal{O}_{X_y}$. So, $P = P_1$ is a prime divisor.

(2) Let $y \notin \Delta$. Note that $P|_{X_y} \sim \mathcal{O}_{X_y}$. So, it suffices to show $X_y \cap P = \mathcal{O}_{X_y}$. If $g^{-1}(y) = y$ after iteration, then $X_y \cap P \subseteq B_{f|_{X_y}} = \mathcal{O}_{X_y}$ by Lemma 2.20. If y is not g^{-1}-periodic, there exists some $y' \in U$ with $g^s(y') = y$ for some $s > 0$. Note that $f^s(X_{y'} \cap P) = X_y \cap P$. Then $X_y \cap P = \mathcal{O}_{X_y}$ by Lemma 2.20.

(3) Note that $X_y \cap P$ has six irreducible components. So, $\deg \sigma = 6$ and hence $\sharp B_\sigma \geqslant 2$ by the simply connectedness of $\mathbb{P}^1$ and $\mathbb{A}^1$.

Lemma 6.7 is proved. □

Proof of Theorem 6.6. We adopt Lemma 6.7 and use its notation there. The proof is similar to the proof of Theorem 6.3. We rewrite it here for the convenience of the reader.

By Lemma 6.7 (2), $(K_X + P)|_{X_y} \sim 0$ for any $y \notin \Delta$. Since π is a K_X-negative contraction of relative Picard number one, due to the cone theorem [28, Theorem 3.7], we have $K_X + P \in \pi^* \mathrm{Pic}(Y)$. By Corollary 2.12, it follows that $(X, P + X_y)$ is lc for any $y \in Y$. Let $y_0 \in B_{\pi|_P} = B_\sigma \neq \emptyset$. Then we obtain a contradiction to Corollary 2.7. □

7 | TWO DIVISORIAL CONTRACTIONS, PROOF OF THEOREM 7.1

In this section, we treat $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}_{\mathrm{bir}}$ (cf. Notation 3.1) where τ is a divisorial contraction. We aim to show:

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Theorem 7.1. *There is no* $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}_{\mathrm{bir}}$ *with* τ *being divisorial.*

Proof. Suppose the contrary that there exists $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z) \in \mathcal{S}_{\mathrm{bir}}$ with τ being divisorial. Denote by E (resp., F) the π-exceptional (resp., τ-exceptional) divisor. Note that the τ-exceptional curve must be E-positive; otherwise, the divisor $-E$ would be nef, which is absurd. In particular, we have $E \neq F$ by [28, Lemma 3.39] and the pseudo-effective cone $\mathrm{PE}(X)$ is generated by E and F. Note that $-K_X$ is big because X is of Fano type. So, there exist coprime positive integers x_1, x_2, x_3 such that

$$x_1 K_X + x_2 E + x_3 F \equiv 0.$$

We try to derive a contradiction based on this.

Claim 7.2. The intersection $E \cap F$ dominates both $\pi(E)$ and $\tau(F)$.

Proof. If $E \cap F$ does not dominate $\tau(F)$, then E is τ-trivial and hence $E = \tau^*\tau(E)$ by the cone theorem [28, Theorem 3.7]. Note that $\tau(E)$ is an ample $\mathbb{Q}$-Cartier divisor because $\rho(Z) = 1$ and Z is $\mathbb{Q}$-factorial (see [28, Proposition 3.36]). So, E is nef, a contradiction to [28, Lemma 3.39] again. Similarly, we have that $E \cap F$ dominates $\pi(E)$. □

Claim 7.3. We have $\dim \pi(E) = 0$ and $\dim \tau(F) = 1$. Moreover, the contraction π is of type E3, E4 or E5.

Proof. By Lemma 3.4, the contraction π is not of type E2.

Suppose π is of type E1. It follows from Theorem 2.21 that Y is a smooth Fano threefold of Picard number 1 and π is the blowup of Y along a smooth curve $\pi(E)$. Then $Y \cong \mathbb{P}^3$ by [2] or [20]. Denote by $g : Y \to Y$ the induced endomorphism. As $\pi(F)$ is a g^{-1}-invariant prime divisor, by [21] and [46, section 5], it is linear. Note that $E \cap F$ dominates $\pi(E)$ by Claim 7.2. It follows that $\pi(F)$ contains a g^{-1}-invariant curve $\pi(E)$. By [19, Theorem], we see that $\pi(E)$ is linear as well. Then X is the blowup of $\mathbb{P}^3$ along a line ℓ. In this case, the one-dimensional family of planes passing through the given line ℓ gives rise to a $\mathbb{P}^2$-bundle over $\mathbb{P}^1$ after the blowup, a contradiction to the setup of τ being divisorial.

Suppose $\tau(F)$ is a point. Then $\pi(E)$ has to be a curve and thus π is of type E1, for otherwise, π and τ would contract some common curve. However, this is impossible as we argued above. □

By Claim 7.3 and Theorem 2.21, the divisor E is isomorphic to either $\mathbb{P}^2$ or an irreducible (possibly singular) quadric surface. Let C_π be a line in E. Then $K_X \cdot C_\pi = -1$.

Recall that f is q-polarized (cf. [36, Corollary 3.12]). Denote by Δ_E (resp., Δ_F, Δ_{E+F}) the divisor $R_f - (q-1)E$ (resp., $R_f - (q-1)F$, $R_f - (q-1)(E+F)$).

Claim 7.4. The divisors $-(K_X + E + F)$ and Δ_{E+F} are big.

Proof. By the logarithmic ramification divisor formula (see [23, Theorem 11.5]), we have

$$f^*(-(K_X + E + F)) - (-(K_X + E + F)) = \Delta_{E+F}.$$

Hence, Δ_{E+F} is an effective divisor with $E \nsubseteq \operatorname{Supp} \Delta_{E+F}$ and $F \nsubseteq \operatorname{Supp} \Delta_{E+F}$. Recall that $\mathrm{PE}(X)$ is generated by the two exceptional divisors E and F. So, a divisor is big if and only if it is numeri-

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cally equivalent to a positive combination of these two divisors. By [33, Theorem 1.1], we see that $-(K_X + E + F)$ is big if Δ_{E+F} is big.

Suppose the contrary that Δ_{E+F} is not big. Then either $\Delta_{E+F} \sim_{\mathbb{Q}} \lambda E$ or $\Delta_{E+F} \sim_{\mathbb{Q}} \lambda F$ for some integer $\lambda \geqslant 0$. Note that $\kappa(X, E) = \kappa(X, F) = 0$. This forces $\lambda = 0$ and thus

$$K_X + E + F = f^*(K_X + E + F).$$

By [33, Theorem 1.1], we have $K_X + E + F \equiv 0$. From [41, Theorem 1.5], we then obtain a toric pair $(X, E + F)$. However, the boundary divisor of a smooth projective toric threefold should have at least 4 irreducible components (cf. [37, Remark 4.6]). So, we get a contradiction. □

Claim 7.5. We have that $x_1 < x_2$ and $x_1 < x_3$.

Proof. Suppose $x_1 \geqslant x_2$. Then we have the reconfigured equation

$$(x_1 - x_2)K_X + x_2(K_X + E) + x_3F \equiv 0$$

and its f-pullback

$$(x_1 - x_2)(K_X - R_f) + x_2(K_X + E - \Delta_E) + x_3qF \equiv 0.$$

Subtracting the above two equations from each other, we have

$$(x_1 - x_2)R_f + x_2\Delta_E \equiv x_3(q - 1)F.$$

This contradicts Claim 7.4, noting that $x_2 > 0, \Delta_E > \Delta_{E+F}$ is big, and F is an extremal ray in $\mathrm{PE}(X)$. So, we have $x_1 < x_2$ and similarly $x_1 < x_3$. □

Claim 7.6. We have $K_X \cdot C_\tau \geqslant 0$ for any curve C_τ contracted by τ.

Proof. Suppose the contrary that $K_X \cdot C_\tau < 0$. Then $-K_X$ is ample and we get a contradiction to Theorem 3.5. □

Claim 7.7. Let C_τ be a curve contracted by τ. Then $E \cdot C_\tau > K_X \cdot C_\tau$ and $E \cdot C_\tau > 1$.

Proof. By Claims 7.3, 7.6 and Theorem 2.21, we have

$$0 \leqslant K_X \cdot C_\tau \leqslant K_Y \cdot \pi_* C_\tau + E \cdot C_\tau < E \cdot C_\tau.$$

Suppose the contrary that $E \cdot C_\tau = 1$. Then $K_X \cdot C_\tau = 0$. The equation

$$0 = (x_1K_X + x_2E + x_3F) \cdot C_\tau = x_2 + x_3F \cdot C_\tau$$

implies $\frac{x_2}{x_3} \in \mathbb{Z}$ and hence the equation

$$0 = \left(\frac{x_1}{x_3}K_X + \frac{x_2}{x_3}E + F\right) \cdot C_\pi = -\frac{x_1}{x_3} + \left(\frac{x_2}{x_3}E + F\right) \cdot C_\pi$$

implies $\frac{x_1}{x_3} \in \mathbb{Z}$. So, $x_1 \geqslant x_3$, a contradiction to Claim 7.5. □

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Claim 7.8. There exists a curve C_τ contracted by τ such that $K_X \cdot C_\tau = 0$, $E \cdot C_\tau = 2$ and $F \cdot C_\tau = -2$.

Proof. Denote by $h := f|_Z$. By Claims 7.2 and 7.3, we see that $\tau(F)$ is a (possibly singular) curve and $E \cap F$ dominates $\tau(F)$. Denote by S the union of one-dimensional components of $\mathrm{Sing}(F)$. By Lemma 2.10, we have $f^{-1}(S) = S$ after iteration. By Lemma 2.9, the divisor $E + F$ has simple normal crossing near the generic points of $E \cap F$. So, there exists a finite subset $T \subsetneq F$ such that $E|_{F\setminus T}$ is a smooth divisor. By the generic smoothness, there exists a nonempty open subset $U \subseteq \tau(F)\setminus\tau(T)$ such that

- $\mathrm{Sing}(F_z) = S \cap F_z$,
- $E \cap S \cap F_z = \emptyset$, and
- $E|_{F_z}$ is smooth, that is, the sum of reduced points

for any $z \in U$. By [12, Theorem 5.1], the set of periodic points $\mathrm{Per}(h|_{\tau(F)})$ is Zariski dense in $\tau(F)$. Take $z \in \mathrm{Per}(h|_{\tau(F)}) \cap U$ and C_τ any irreducible component of F_z. After iteration, we may assume $h(z) = z$ and $f(C_\tau) = C_\tau$. Note that C_τ admits at most two $(f|_{C_\tau})^{-1}$-periodic points. Then

$$\sharp E \cap C_\tau + \sharp S \cap C_\tau \leqslant 2.$$

Note that $E \cdot C_\tau = \sharp E \cap C_\tau \geqslant 2$ by Claim 7.7. It follows that $E \cdot C_\tau = 2$ and $S \cap C_\tau = \emptyset$. In particular, $F_z = X_z$ is a smooth rational curve and F is smooth around F_z.

Suppose $K_X \cdot C_\tau \geqslant 1$. Then $K_X \cdot C_\tau = 1$ by Claim 7.7. Recall that $K_X \cdot C_\pi = -1$. If $E \cdot C_\pi = -2$, then $E \equiv 2K_X$, a contradiction to the bigness of $-K_X$. By Claim 7.3 and Theorem 2.21, we have $E \cdot C_\pi = -1$. Intersecting $x_1K_X + x_2E + x_3F \equiv 0$ with C_π and C_τ, we have equations:

$$\begin{cases} x_1 + x_2 &= (F \cdot C_\pi)x_3 \\ x_1 + 2x_2 &= -(F \cdot C_\tau)x_3 \end{cases}$$

In particular, we obtain

$$x_1 = (2F \cdot C_\pi + F \cdot C_\tau)x_3$$

and hence $x_3 \mid x_1$, a contradiction to Claim 7.5. So, $K_X \cdot C_\tau = 0$.

By the adjunction formula, we have

$$(K_X + F) \cdot C_\tau = K_F \cdot C_\tau = -2$$

where these intersections make sense because we calculate them inside the smooth locus of F. So, $F \cdot C_\tau = -2$. □

End of the proof of Theorem 7.1. By Claim 7.8, we have equations:

$$\begin{cases} x_1 - (E \cdot C_\pi)x_2 &= (F \cdot C_\pi)x_3 \\ 2x_2 &= 2x_3 \end{cases}$$

Then $x_1 = (F \cdot C_\pi + E \cdot C_\pi)x_2$. This contradicts Claim 7.5. □

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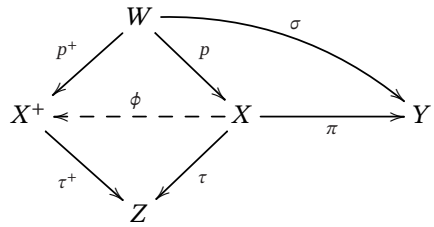
8 | ONE DIVISORIAL AND ONE SMALL CONTRACTION

In this section, we deal with $(X \circlearrowleft f, Y \overset{\pi}{\leftarrow} X \overset{\tau}{\to} Z) \in \mathcal{S}_{\mathrm{bir}}$ (cf. Notation 3.1) where τ is a small contraction. We aim to show:

Theorem 8.1. *There is no $(X \circlearrowleft f, Y \overset{\pi}{\leftarrow} X \overset{\tau}{\to} Z) \in \mathcal{S}_{\mathrm{bir}}$ with τ being a small contraction.*

To prove Theorem 8.1, we shall stick to the following notation till the end of this section.

Notation 8.2. Let $(X \circlearrowleft f = f_X, Y \overset{\pi}{\leftarrow} X \overset{\tau}{\to} Z) \in \mathcal{S}_{\mathrm{bir}}$ with τ being small. Then we have the following commutative diagram:



where

- the map $\phi : X \dashrightarrow X^+$ is the log flip induced by the log flipping contraction τ,
- the variety W is the normalization of the graph of ϕ with two projections $p : W \to X$ and $p^+ : W \to X^+$,
- the commutative diagram is f-equivariant: precisely, we have the induced endomorphisms f_Y, f_Z, f_{X^+} by [36, Remark 6.3 ∼ Lemma 6.6] and f_W by the universal property of the graph and normalization, and
- f_Y is q-polarized because $\rho(Y) = 1$ and thus all the induced endomorphisms are q-polarized by [36, Corollary 3.12].

Moreover, we introduce the following notation as well as fundamental facts.

(1) Denote by E the π-exceptional divisor and by C_τ a fixed τ-exceptional irreducible curve. After iteration, we may assume that C_τ is f^{-1}-invariant.
(2) Let $E_W := p^{-1}(E) = p^*(E)$ which is Cartier and denote by F_W a prime divisor contained in $p^{-1}(C_\tau)$ which dominates C_τ. After iteration, we may also assume that both E_W and F_W are f_W^{-1}-invariant.
(3) Denote by T_{f_Y} (resp., T_{f_W}) the union of f_Y^{-1}-periodic (resp., f_W^{-1}-periodic) prime divisors. By [38, Corollary 3.8], both T_{f_W} and T_{f_Y} have only finitely many prime divisors. After iteration, we may assume every f_Y^{-1}-periodic (resp., f_W^{-1}-periodic) prime divisor is f_Y^{-1}-invariant (resp., f_W^{-1}-invariant). In this way, we may assume

$$(f_-)^*|_{\mathrm{N}^1(-)} = q\,\mathrm{id}_{\mathrm{N}^1(-)}$$

where $-$ takes X, X^+, Y, Z, W.

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8.1 | Intersection of exceptional loci

In this subsection, we prove the following theorem.

Theorem 8.3. *Under Notation 8.2, we have the intersection $E \cdot C_\tau \leqslant 2$ and $\sharp E \cap C_\tau = E \cdot C_\tau$ if C_τ is smooth.*

Lemma 8.4. *Under Notation 8.2, for each $n \in \mathbb{Z}_+$, there exists some Weil $\mathbb{Q}$-divisor $D_{Y,n}$ such that the pair $(Y, (1-\frac{1}{n})T_{f_Y} + D_{Y,n})$ is klt log Calabi–Yau (cf. Definition 2.2).*

Proof. Recall that Theorem 2.21 implies Y is $\mathbb{Q}$-factorial. By Lemma 2.8, the pair (Y, T_{f_Y}) is lc and $-(K_Y + T_{f_Y})$ is pseudo-effective. Hence, $(Y, (1-\frac{1}{n})T_{f_Y})$ is klt and $-(K_Y + (1-\frac{1}{n})T_{f_Y})$ is ample, noting that $\rho(Y) = 1$. Let $A \equiv -m(K_Y + (1-\frac{1}{n})T_{f_Y})$ be an integral very ample divisor for a sufficiently divisible $m = m(n)$. It follows from Bertini's theorem that a general member $H \in |A|$ intersects T_{f_Y} transversally. Therefore, the pair $(Y, (1-\frac{1}{n})T_{f_Y} + \frac{1}{m}H)$ is klt log Calabi–Yau. □

Notation 8.5. We follow Notation 8.2. By Lemma 8.4, we have a series of klt log Calabi–Yau pairs $(Y, (1-\frac{1}{n})T_{f_Y} + D_{Y,n})$ for each $n \in \mathbb{Z}^+$. Let $D_{W,n}$ be the Weil $\mathbb{Q}$-divisor on W obtained by

$$K_W + \Big(1-\frac{1}{n}\Big)T_{f_W} + D_{W,n} = \sigma^*\Big(K_Y + \Big(1-\frac{1}{n}\Big)T_{f_Y} + D_{Y,n}\Big) \equiv 0.$$

The following construction is a special application of f-pairs as in Yoshikawa [52, Proposition 6.2]. For each $i \geqslant 0$, denote by

$$D_{Y,n,i} := \frac{1}{(\deg f_Y)^i}(f_Y^i)_*\Big(R_{f_Y^i} + \Big(1-\frac{1}{n}\Big)T_{f_Y} + D_{Y,n}\Big) - T_{f_Y},$$

$$D_{W,n,i} := \frac{1}{(\deg f_W)^i}(f_W^i)_*\Big(R_{f_W^i} + \Big(1-\frac{1}{n}\Big)T_{f_W} + D_{W,n}\Big) - \Big(1-\frac{1}{n}\Big)T_{f_W}.$$

Lemma 8.6. *Under Notation 8.5, For any $n \in \mathbb{N}$, there exists $i_n \gg 1$ such that the pair $(W, (1-\frac{1}{n})T_{f_W} + D_{W,n,i_n})$ is klt log Calabi–Yau.*

Proof. By [13, Lemma 1.1] and its proof, the pair $(Y, (1-\frac{1}{n})T_{f_Y} + D_{Y,n,i})$ is klt log Calabi–Yau.

Claim 8.7. $K_W + (1-\frac{1}{n})T_{f_W} + D_{W,n,i} = \sigma^*(K_Y + (1-\frac{1}{n})T_{f_Y} + D_{Y,n,i}) \equiv 0$.

Proof. Recall that $f_W^*|_{\mathrm{N}^1(W)} = q\,\mathrm{id}_{\mathrm{N}^1(W)}$. Then

$$\begin{aligned}(f_W^i)^*\Big(K_W + \Big(1-\frac{1}{n}\Big)T_{f_W} + D_{W,n,i}\Big) &\equiv q^i K_W + R_{f_W^i} + \Big(1-\frac{1}{n}\Big)T_{f_W} + D_{W,n}\\ &= (q^i-1)K_W + R_{f_W^i} \equiv 0.\end{aligned}$$

This implies that $K_W + (1-\frac{1}{n})T_{f_W} + D_{W,n,i} \equiv 0$. Therefore, we have

$$\Delta := K_W + \Big(1-\frac{1}{n}\Big)T_{f_W} + D_{W,n,i} - \sigma^*\Big(K_Y + \Big(1-\frac{1}{n}\Big)T_{f_Y} + D_{Y,n,i}\Big) \equiv 0.$$

Note that $\mathrm{Supp}\,\Delta \subseteq \mathrm{Exc}(\sigma)$ since $\sigma_*\Delta = 0$. So, $\Delta = 0$ by [28, Lemma 3.39]. □

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Claim 8.8. There exists some $i_n \geqslant 0$ such that D_{W,n,i_n} is effective.

Proof. We refer to [52, Claim 6.3] for the original proof. Since our pair here is slightly different, we provide a proof for the reader's convenience.

Note that $D_{Y,n,i}$ is effective and $\sigma_* D_{W,n,i} = D_{Y,n,i}$ for each $i > 0$. Hence, it suffices to show that $\mathrm{mult}_P D_{W,n,i} \geqslant 0$ for $i \gg 1$, where P is an arbitrary exceptional prime divisor of $\sigma : W \to Y$. We may assume P is f_W^{-1}-invariant after iteration. Write $f_W^* P = rP$ and $(f_W)_* P = eP$. Since $(Y, (1 - \frac{1}{n})T_{f_Y} + D_{Y,n})$ is klt, we have $a := \mathrm{mult}_P((1 - \frac{1}{n})T_{f_W} + D_{W,n}) < 1$ and

$$\begin{aligned}
\mathrm{mult}_P D_{W,n,i} &= \mathrm{mult}_P \left(\frac{f_{W*}^i \left(R_{f_W^i} + \left(1 - \frac{1}{n}\right) T_{f_W} + D_{W,n} \right)}{\deg f_W^i} - \left(1 - \frac{1}{n}\right) T_{f_W} \right) \\
&= \frac{1}{r^i}(r^i - 1 + a) - \left(1 - \frac{1}{n}\right) \mathrm{mult}_P T_{f_W} \\
&= \frac{1}{n} + \frac{a-1}{r^i} \geqslant 0
\end{aligned}$$

for $i \gg 1$. Our claim is thus proved. □

Back to the proof of Lemma 8.6, from Claims 8.8 and 8.7, we obtain a klt log Calabi–Yau pair $(W, (1 - \frac{1}{n})T_{f_W} + D_{W,n,i_n})$, which concludes our proof. □

Lemma 8.9. *Under Notation 8.2, there exists a hypersurface H of W such that the following hold.*

(1) *The hypersurface H does not contain any irreducible component of $E_W \cap F_W$ and* $\mathrm{Sing}(W)$.
(2) *The pair $(H, E_H + F_H)$ is lc where $E_H = E_W \cap H = E_W|_H$ is Cartier and $F_H = F_W \cap H$.*

Proof. Note that there are only finitely many irreducible components of $E_W \cap F_W$ and $\mathrm{Sing}(W)$. So, we can choose a general hypersurface H such that H does not pass through any generic points of those irreducible components which gives (1).

By [49, Theorem 7], we can further choose H general such that H is normal. Let $U_W = W \setminus (\mathrm{Sing}(W) \cup (E_W \cap F_W))$ and $U_H = H \cap U_W$. Note that both U_W and U_H are smooth.

We follow Notation 8.5 and apply Lemma 8.6. Let $i_n \gg 1$. Since $E_W, F_W \subseteq T_{f_W}$, we write

$$\left(1 - \frac{1}{n}\right) T_{f_W} + D_{W,n,i_n} = \left(1 - \frac{1}{n}\right)(E_W + F_W) + \Delta_{W,n}$$

where $\Delta_{W,n}$ is an effective Weil $\mathbb{Q}$-divisor. Note that $(W, (1 - \frac{1}{n})(E_W + F_W) + \Delta_{W,n})$ is klt log Calabi–Yau. Then $(W, (1 - \frac{1}{n})(E_W + F_W) + \Delta_{W,n} + H)$ is lc by [28, Lemma 5.17(2)]. By [27, Theorem], for each n, we have

$$K_H + B_n = \left(K_W + \left(1 - \frac{1}{n}\right)(E_W + F_W) + \Delta_{W,n} + H\right)|_H,$$

where B_n is the Shokurov's different of $(1 - \frac{1}{n})(E_W + F_W) + \Delta_{W,n}$ such that (H, B_n) is lc. By the (smooth) adjunction formula for $U_H \subseteq U_W$, we have

$$K_{U_H} + \left(1 - \frac{1}{n}\right)(E_{U_H} + F_{U_H}) + \Delta_{U_H,n} = \left(K_{U_W} + \left(1 - \frac{1}{n}\right)(E_{U_W} + F_{U_W}) + \Delta_{U_W,n} + H_{U_W}\right)|_{U_H}$$

where $\Delta_{U_-,n} := \Delta_{W,n}|_{U_-}$, $E_{U_-} := E_W|_{U_-}$ and similarly for F_{U_-}, H_{U_W}.

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The above two formulae imply that

$$B_n|_{U_H} = \Big(1-\frac{1}{n}\Big)(E_{U_H}+F_{U_H})+\Delta_{U_H,n} \geqslant \Big(1-\frac{1}{n}\Big)(E_{U_H}+F_{U_H}).$$

Note that $\dim W\backslash U_W \leqslant 1$ and $\dim H\backslash U_H \leqslant 0$ by (1). Then B_n is the extension divisor of $(1-\frac{1}{n})(E_{U_H}+F_{U_H})+\Delta_{U_H,n}$. Since E_{U_H} and F_{U_H} are integral divisors, they are reduced divisors by noting that $B_n \leqslant 1$. Therefore, $E_H = E_W \cap H = E_W|_H$ and $F_H = F_W \cap H$ are the extension divisors of E_{U_H} and F_{U_H}, respectively. In particular, $(1-\frac{1}{n})(E_H+F_H) \leqslant B_n$. So, $(H,(1-\frac{1}{n})(E_H+F_H))$ is numerically lc and hence lc; see [16, Proposition 3.5 (2)]. By taking $n\to\infty$, we conclude that (H, E_H+F_H) is lc. □

Remark 8.10. We are not able to apply Lemma 2.8 directly because $K_W+E_W+F_W$ is not known to be $\mathbb{Q}$-Cartier.

The following lemma is a slight extension of Lemma 2.9.

Lemma 8.11. *Under the notation in Lemma 8.9, the pair (H, E_H+F_H) is simple normal crossing near $E_H\cap F_H$. In particular, the restriction divisor $E_H|_{F_H}$ is reduced.*

Proof. Let $e\in E_H\cap F_H$ be a point. Note that (H,E_H) is numerically dlt (and hence dlt) at $E_H\cap F_H$ (cf. [28, Corollary 4.2 and Proposition 4.11]). By Lemma 8.9, E_H is Cartier. By [28, Corollary 5.56], H is smooth near $E_H\cap F_H$. By [28, Theorem 4.15(1)], E_H+F_H has exactly two analytic branches at e. In particular, E_H and F_H are smooth at e. Suppose E_H intersects F_H not transversally at e. By blowing up H along e, we obtain an lc pair with the boundary divisor a union of three curves passing through one point, a contradiction. □

Proof of Theorem 8.3. We follow Lemma 8.9 and its notation therein. Consider the following commutative diagram:

$$\begin{array}{ccccc} & & \widetilde{F_H} & \xrightarrow{n_{F_H}} & F_H \\ & & \downarrow{\scriptstyle \widetilde{p_F}} & & \downarrow{\scriptstyle p_F} \\ f_{\widetilde{C_\tau}} \circlearrowright & & \widetilde{C_\tau} & \xrightarrow{n_{C_\tau}} & C_\tau \quad \circlearrowleft f_{C_\tau} \end{array}$$

where

- n_{F_H} and n_{C_τ} are normalizations of F_H and C_τ, respectively.
- p_F is the restriction of p on F_H and induces $\widetilde{p_F}$.
- f_{C_τ} is the restriction of $f_X : X\to X$ on C_τ and induces $f_{\widetilde{C_\tau}}$.

Recall from Notation 8.2 (1) that, the chosen component C_τ is f^{-1}-invariant. Since $H|_{F_W}$ is ample in F_W with support $F_H = F_W\cap H$, we have p_F is surjective. Observe that $\operatorname{Supp} n_{C_\tau}^*(E|_{C_\tau}) = n_{C_\tau}^{-1}(E\cap C_\tau)$ is $(f_{\widetilde{C_\tau}})^{-1}$-invariant and there are at most two $(f_{\widetilde{C_\tau}})^{-1}$-periodic points on $\widetilde{C_\tau}$, so $\sharp\operatorname{Supp} n_{C_\tau}^*(E|_{C_\tau}) \leqslant 2$.

Suppose $E\cdot C_\tau \geqslant 3$. Then there exists a point $e_1\in\widetilde{C_\tau}$ such that $\operatorname{mult}_{e_1} n_{C_\tau}^*(E|_{C_\tau}) \geqslant 2$. Let $e_0 = n_{C_\tau}(e_1)\in E\cap C_\tau$ and choose $e_3\in\widetilde{F_H}$ such that $\widetilde{p_F}(e_3) = e_1$. Let $e_2 = n_{F_H}(e_3)\in E_H\cap F_H$. By

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Lemma 8.11, H is smooth at e_2 and hence n_{F_H} is isomorphic at e_3. So, we have

$$\mathrm{mult}_{e_2} E_H|_{F_H} = \mathrm{mult}_{e_3} n_{F_H}^*(E_H|_{F_H}) = \mathrm{mult}_{e_3}(n_{C_\tau} \circ \widetilde{p_F})^*(E|_{C_\tau}) \geqslant 2,$$

a contradiction to Lemma 8.11.

Suppose C_τ is smooth and $\sharp E \cap C_\tau < E \cdot C_\tau$. Then n_{C_τ} is the identity map and there exists a point $e_1 \in \widetilde{C_\tau} = C_\tau$ such that $\mathrm{mult}_{e_1} E|_{C_\tau} \geqslant 2$. Then we obtain the same contradiction. □

8.2 | Proof of Theorem 8.1

We first prepare the following general lemma.

Lemma 8.12. *Let X be a projective variety and D a base-point-free Cartier divisor on X. For any singular curve C on X with $D \cdot C > 0$, we then have $D \cdot C \geqslant 2$.*

Proof. Consider the morphism $\iota : X \to \mathbb{P}^N$ induced by the linear system $|D|$ such that $D = \iota^* H$ where H is a hyperplane in $\mathbb{P}^N$. Let W be the image $\iota(X)$. If $\deg \iota|_C \geqslant 2$, then by the projection formula, we have $D \cdot C = (\deg \iota|_C) H|_W \cdot \iota(C) \geqslant 2$. If $\deg \iota|_C = 1$, then C is birational onto $\iota(C)$ and hence $\iota(C)$ is singular as well, in particular, $\iota(C)$ is nonlinear in $\mathbb{P}^N$ and thus $D \cdot C = H|_W \cdot \iota(C) = H \cdot \iota(C) \geqslant 2$. □

In the remaining subsection, we prove Theorem 8.1.

Proof of Theorem 8.1. Our first task is to find another totally invariant prime divisor F.

Claim 8.13. We have $K_X \cdot C_\tau \geqslant 0$ and $\dim \pi(E) = 0$. Moreover, the contraction π is of type E3, E4 or E5.

Proof. By the same proof of Claim 7.6, we have $K_X \cdot C_\tau \geqslant 0$. By Lemma 3.4, π is not of type $E2$.

Suppose π is of type E1. Recall the proof of Claim 7.3, we see that $Y \cong \mathbb{P}^3$. By Theorems 2.21 and 8.3, we have

$$0 \leqslant K_X \cdot C_\tau = K_Y \cdot \pi_* C_\tau + E \cdot C_\tau \leqslant -2.$$

This is a contradiction. □

Claim 8.14. We have that $K_X \cdot C_\tau = 0$ and hence X is weak Fano.

Proof. By Claim 8.13, $K_X \cdot C_\tau \geqslant 0$ and π is of type E3, E4 or E5.

Suppose π is of type E5. Then we have

$$0 \leqslant K_X \cdot C_\tau = K_Y \cdot \pi_* C_\tau + \frac{1}{2} E \cdot C_\tau < 1.$$

Therefore, $K_X \cdot C_\tau = 0$.

Suppose π is of type E3 or E4. Then we have

$$K_X \cdot C_\tau = K_Y \cdot \pi_* C_\tau + E \cdot C_\tau.$$

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If $E \cdot C_\tau = 1$, then we still have $K_X \cdot C_\tau = 0$. So, we may assume $E \cdot C_\tau = 2$. By Theorem 8.3, either $\sharp E \cap C_\tau = 2$ or C_τ is singular. Note that $\pi(E)$ is a point. It follows that $\pi(C_\tau)$ is a singular curve. By Theorem 2.21, the divisor $-K_Y$ is base point free. Since $-K_Y$ is ample, by Lemma 8.12, we have $K_Y \cdot \pi_* C_\tau \leqslant -2$. So, $K_X \cdot C_\tau \leqslant 0$. □

By Claim 8.14, we see that τ is a K_X-trivial small contraction and hence it follows from [24, Theorem 1.4.15] (cf. [29, Theorem 2.4], [48]) that there exists a flop $X \dashrightarrow X^+$ over Z such that X^+ is smooth, and K_{X^+} is τ^+-trivial and not pseudo-effective. Then we have the following diagram:

$$\begin{array}{ccccc} X & & \overset{\phi}{- - - - \succ} & & X^+ \\ & {\scriptstyle\pi}\searrow & & \swarrow{\scriptstyle\tau^+} & \\ & & Z & & \\ & {\scriptstyle\tau}\swarrow & & \searrow{\scriptstyle\pi^+} & \\ Y & & & & Y^+ \end{array}$$

where $\pi^+ : X^+ \to Y^+$ is the K_{X^+}-negative contraction. Note that the big divisor $-K_{X^+}$ is τ^+-trivial and π^+-ample. So, X^+ is again weak Fano.

Claim 8.15. We have $(X^+ \circlearrowleft f^+, Y^+ \xleftarrow{\pi^+} X^+ \xrightarrow{\tau^+} Z) \in \mathcal{S}_{\text{bir}}$ with τ^+ being small.

Proof. Note that τ^+ is small. Suppose the contrary that π^+ is a Fano contraction. By Corollary 4.3, we have $\dim Y^+ = 1$. Since ϕ is isomorphic in codimension 1, we have $h^0(X, nE) = h^0(X^+, n\phi(E))$ for each $n > 0$, and hence $\kappa(X^+, \phi(E)) = \kappa(X, E) = 0$, where the second equality is due to E being exceptional. This is a contradiction to Corollary 6.2, noting that X^+ is weak Fano. □

By Claim 8.15 and Theorem 2.21, we see that π^+ is still a divisorial contraction. Now we introduce the following four totally invariant divisors.

• E	the π-exceptional divisor
• E^+	the ϕ-image $\phi(E)$
• F^+	the π^+-exceptional divisor
• F	the ϕ-preimage $\phi^{-1}(F^+)$

Note that E^+ is τ^+-negative and F^+ is π^+-negative. Then $\mathrm{PE}(X) = \phi^*(\mathrm{PE}(X^+))$ is generated by E and F. So, there exists coprime positive integers x_1, x_2, x_3 such that

$$\begin{cases} x_1 K_X + x_2 E + x_3 F & \equiv 0 \\ x_1 K_{X^+} + x_3 F^+ + x_2 E^+ & \equiv 0 \end{cases}$$

We try to derive a contradiction based on this. The strategy is a bit different from that in the proof of Theorem 7.1. The trick here is that the restriction on x_1, x_2, x_3 is coming from both X and X^+ symmetrically.

Note that f is q-polarized. Denote by Δ_E (resp., Δ_F, Δ_{E+F}) the divisor $R_f - (q-1)E$ (resp., $R_f - (q-1)F$, $R_f - (q-1)(E+F)$). By the same proofs of Claims 7.4, 7.5 and 7.7, we have the following three claims. Note that Claim 8.16 is based on $\kappa(X, E) = \kappa(X, F) = 0$, Claim 8.17 is based on Claim 8.16, and Claim 8.18 is based on Claim 8.17.

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Claim 8.16. The divisors $-(K_X + E + F)$ and Δ_{E+F} are big.

Claim 8.17. We have that $x_1 < x_2$ and $x_1 < x_3$.

Claim 8.18. We have that $E \cdot C_\tau \geqslant 2$.

The above induces further restriction.

Claim 8.19. We have that $x_1 = 1$, $x_2 > 2$ and $x_3 = 2$.

Proof. By Claim 8.17, we have $x_3 \geqslant x_1 + 1 \geqslant 2$ and $x_2 \geqslant 2$. By Theorem 8.3 and Claim 8.18, the equation

$$0 = (x_1 K_X + x_2 E + x_3 F) \cdot C_\tau = 2x_2 + (F \cdot C_\tau)x_3$$

implies that $\frac{2x_2}{x_3} \in \mathbb{Z}$ and hence the equation

$$0 = \left(\frac{2x_1}{x_3} K_X + \frac{2x_2}{x_3} E + 2F\right) \cdot C_\pi = -\frac{2x_1}{x_3} + \left(\frac{2x_2}{x_3} E + 2F\right) \cdot C_\pi$$

implies that $\frac{2x_1}{x_3} \in \mathbb{Z}$. By Claim 8.17 again, we have $\frac{2x_1}{x_3} < 2$ which forces $2x_1 = x_3$, and hence $\frac{x_2}{x_1} = \frac{2x_2}{x_3} \in \mathbb{Z}$. By the coprime assumption, we have $x_1 = 1$ and thus $x_3 = 2$ and $x_2 > 1$. Suppose $x_2 = 2$. Then we have

$$0 = (K_X + 2E + 2F) \cdot C_\pi = -1 + 2(E + F) \cdot C_\pi.$$

This is a contradiction. □

End of the proof of Theorem 8.1. By Claim 8.19, we have

$$K_X + x_2 E + 2F \equiv 0.$$

Replacing $(X \circlearrowleft f, Y \xleftarrow{\pi} X \xrightarrow{\tau} Z)$ by $(X^+ \circlearrowleft f^+, Y^+ \xleftarrow{\pi^+} X^+ \xrightarrow{\tau^+} Z)$, we also have

$$K_{X^+} + x_2 F^+ + 2E^+ \equiv 0$$

and hence

$$K_X + 2E + x_2 F = \phi^*(K_{X^+} + x_2 F^+ + 2E^+) \equiv 0.$$

Therefore, $x_2 = 2$, contradicting Claim 8.19. □

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